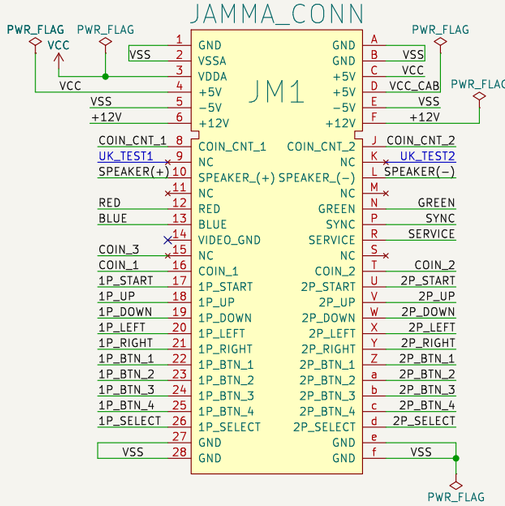
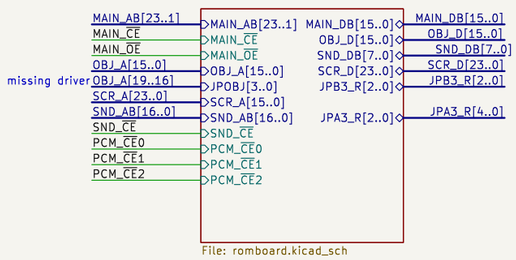
JAMMA_CONN



Capacitors

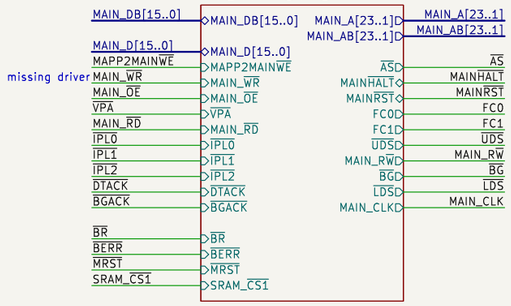
File: capacitors.kicad_sch

ROM board



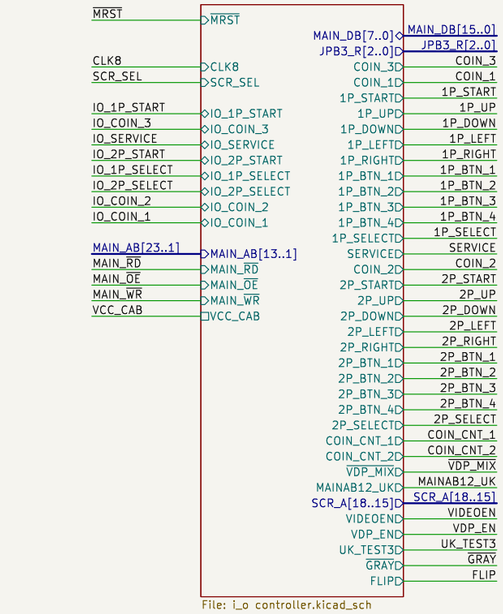
File: romboard.kicad_sch

Main CPU



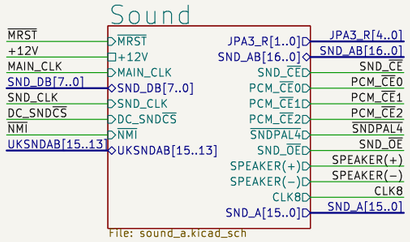
File: main.kicad_sch

IO Controller



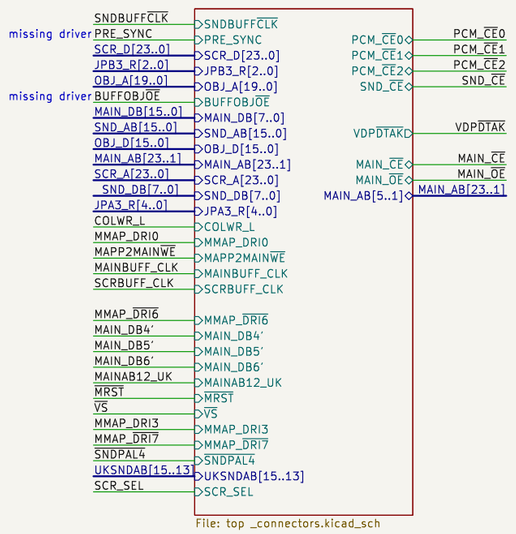
File: i_o controller.kicad_sch

Sound



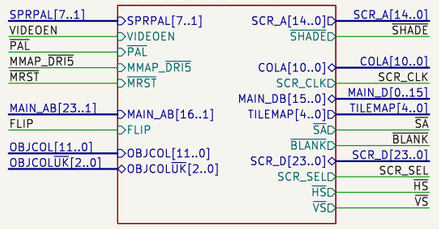
File: sound_a.kicad_sch

Connectors



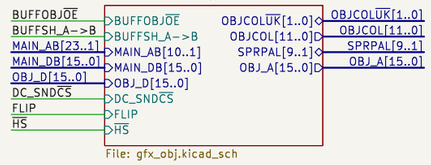
File: top _connectors.kicad_sch

S16B Scroll



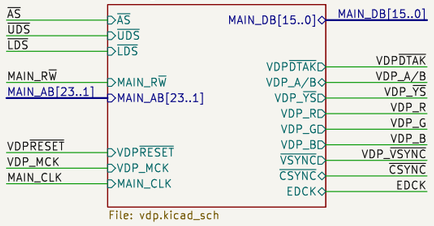
File: gfx_scr.kicad_sch

S16B Ojects



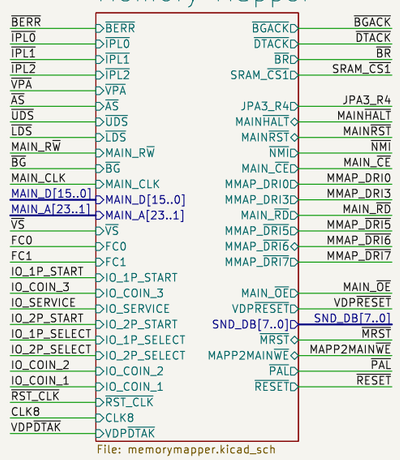
File: gfx_obj.kicad_sch

MegaDrive VDP



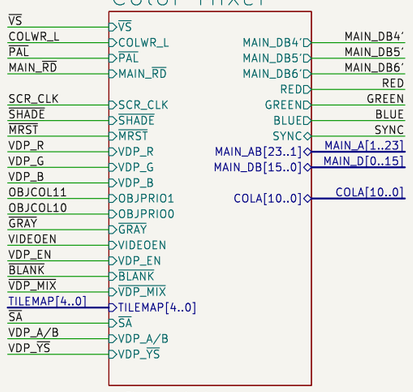
File: vdp.kicad_sch

Memory Mapper



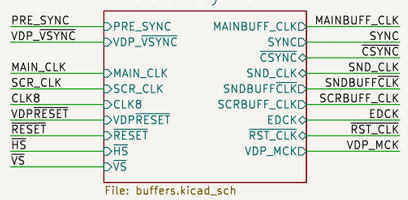
File: memorymapper.kicad_sch

Color Mixer



File: colmix.kicad_sch

Video Sync



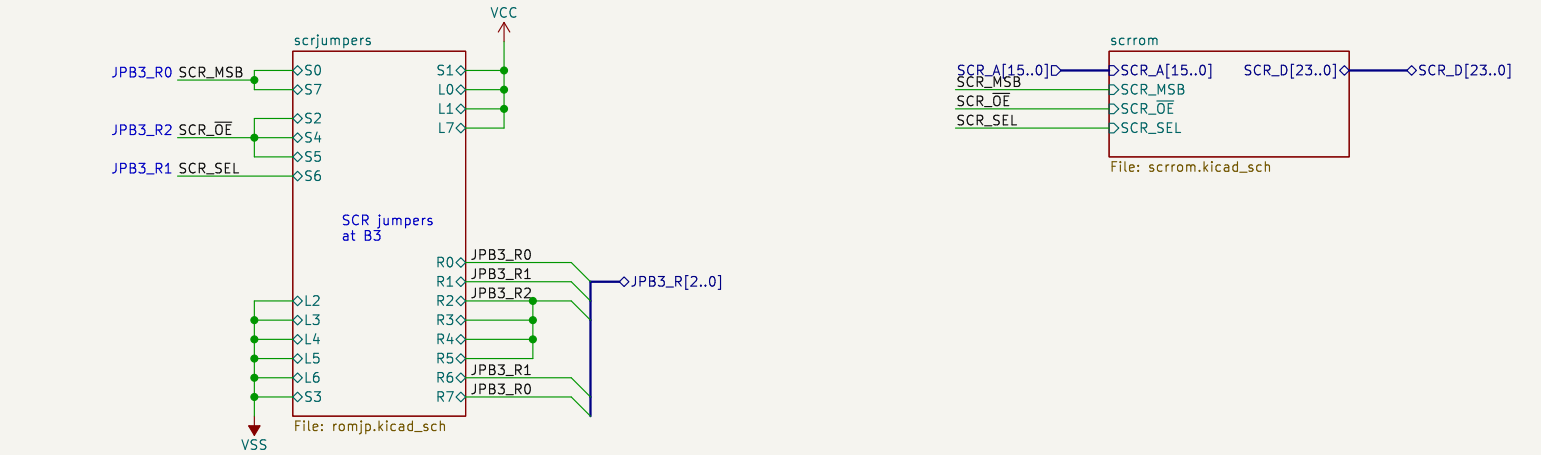
File: buffers.kicad_sch

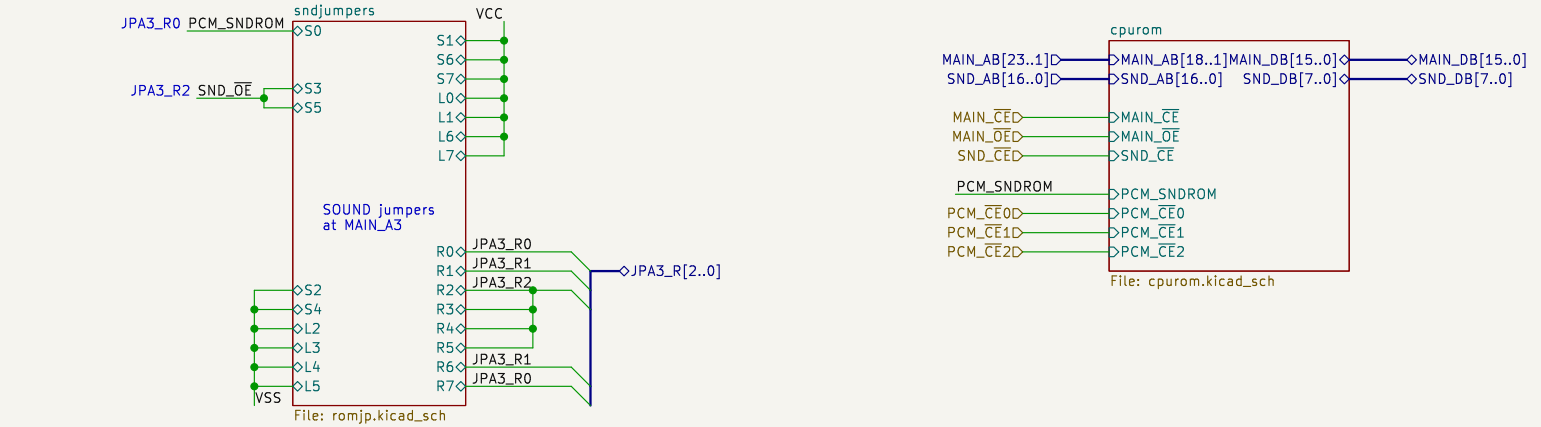
patreon.com/jotego
JOTEGO

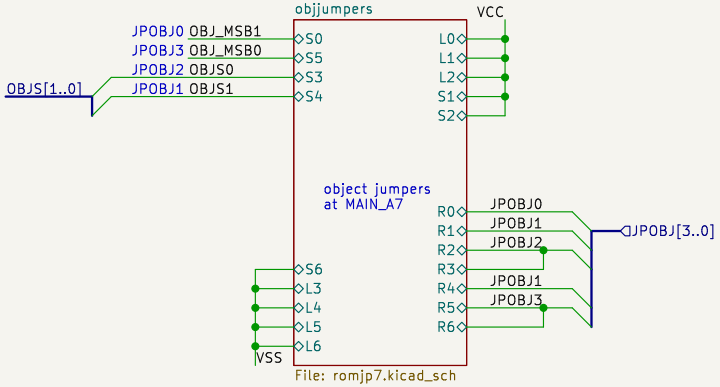
Sheet: /
File: s18.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A3	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 1/26









ROM board

patreon.com/jotego

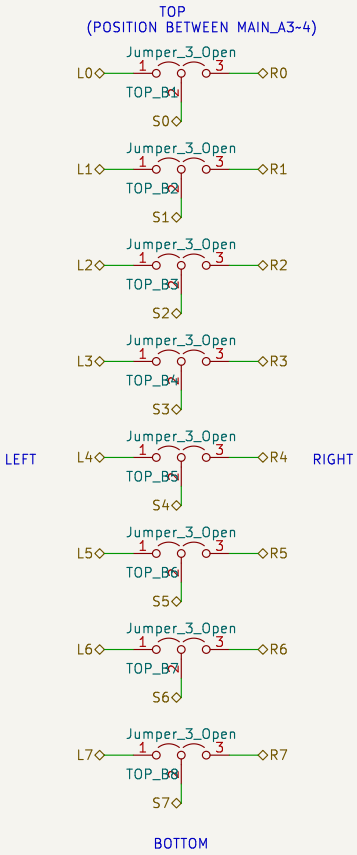
JOTEGO

Sheet: /ROM board/

File: romboard.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 2/26

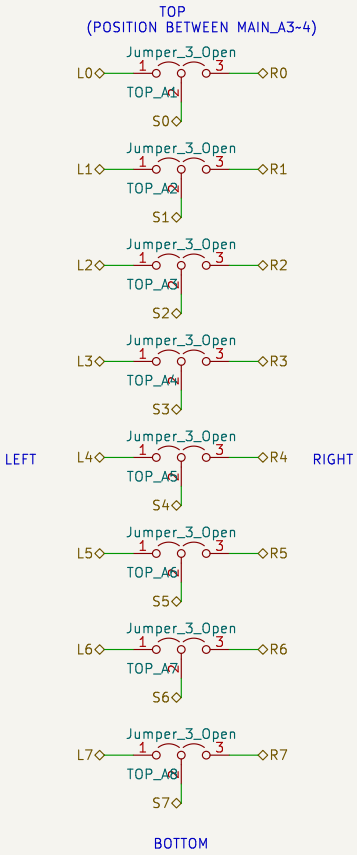


patreon.com/jotego
JOTEGO

Sheet: /ROM board/scrjumpers/
File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 3/26

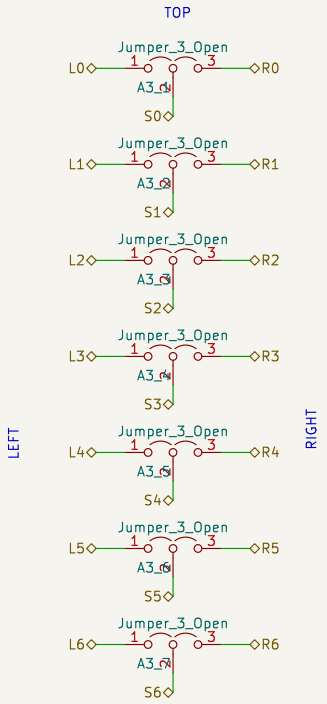


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JOTEGO

Sheet: /ROM board/sndjumpers/
File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 4/26



Seven jumpers

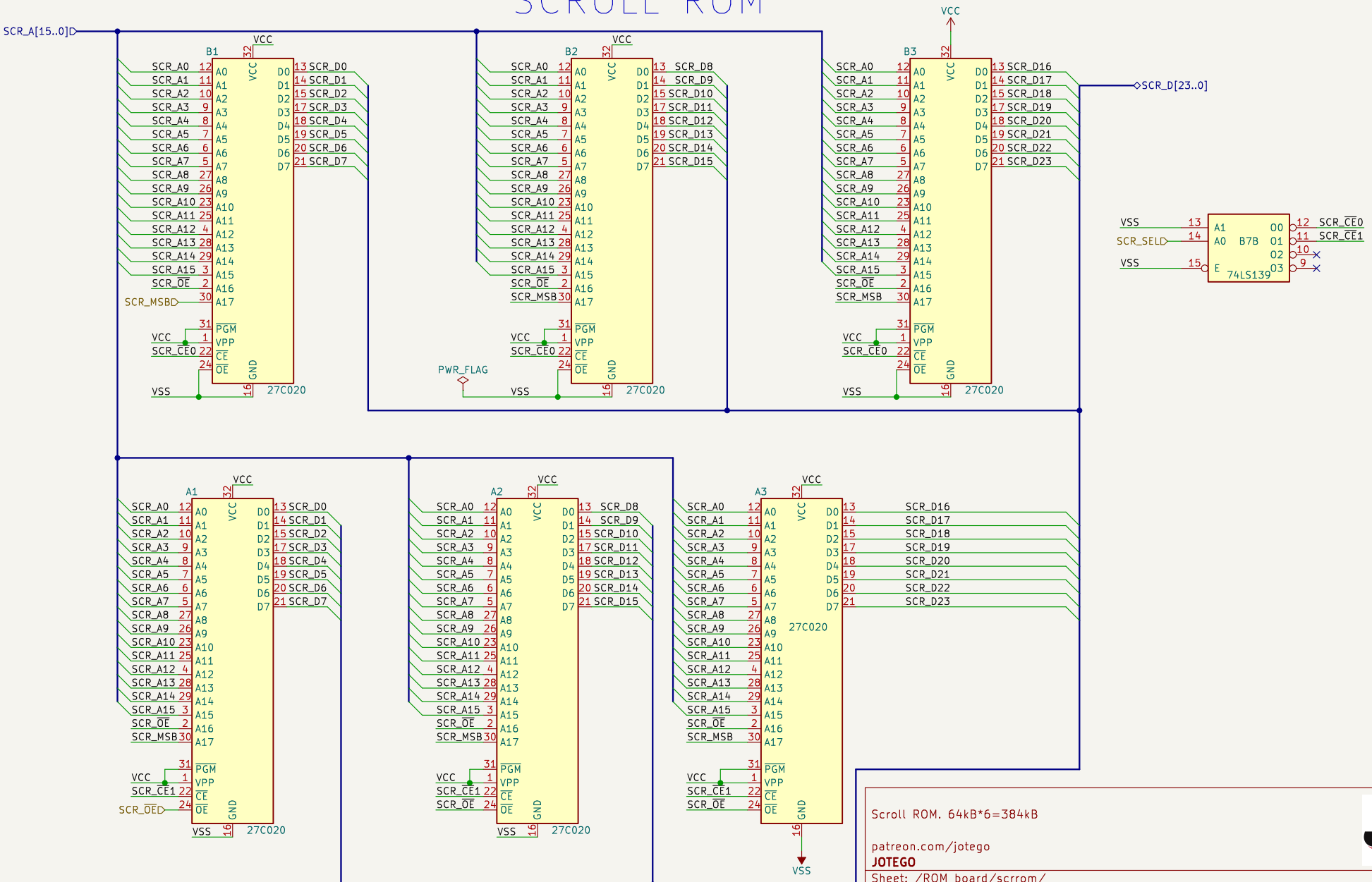
patreon.com/jotego
JOTEGO

Sheet: /ROM board/objjumpers/
File: romjp7.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 5/26

SCROLL ROM



Scroll ROM. 64kB*6=384kB

patreon.com/jotego

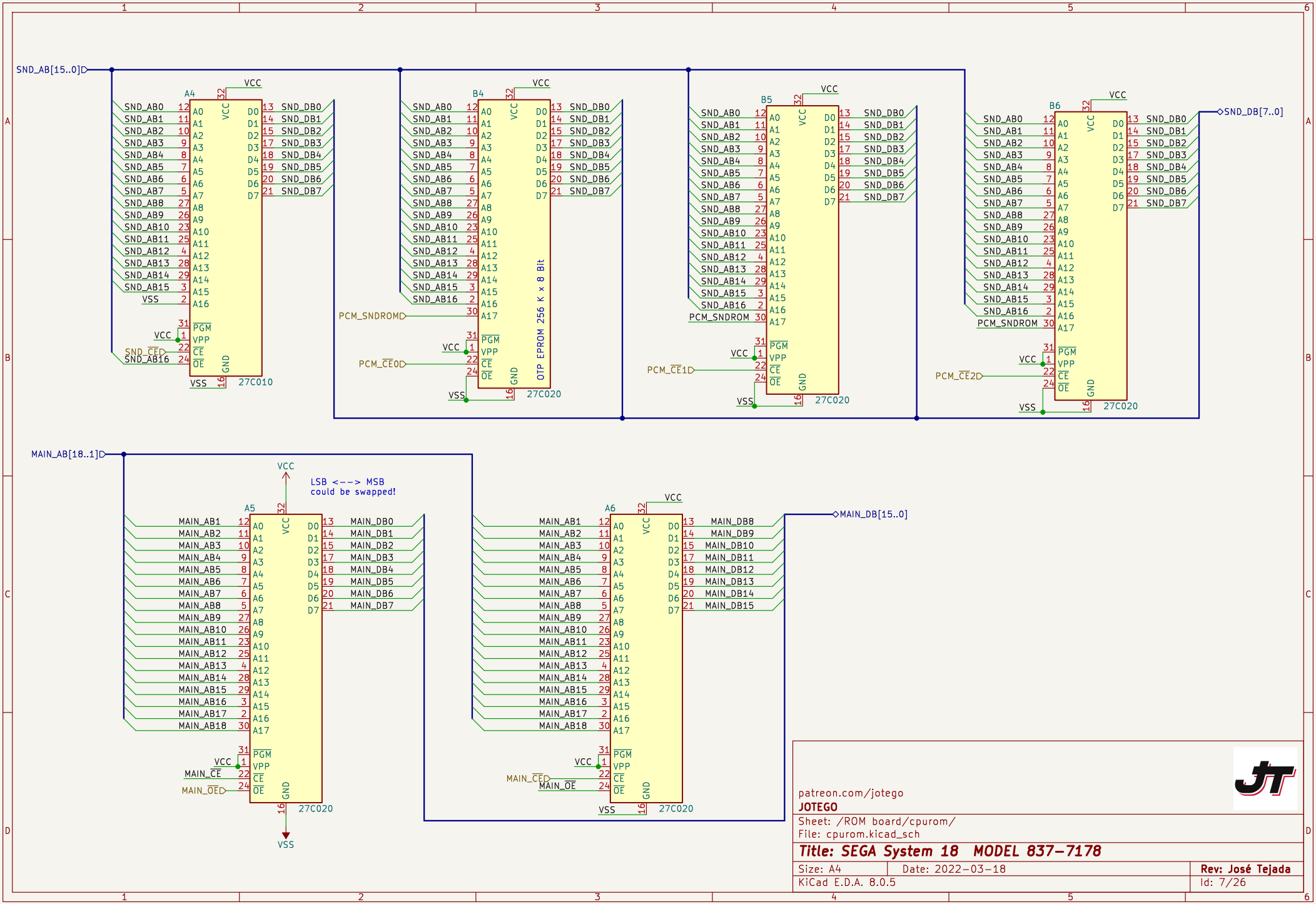
JOTEGO

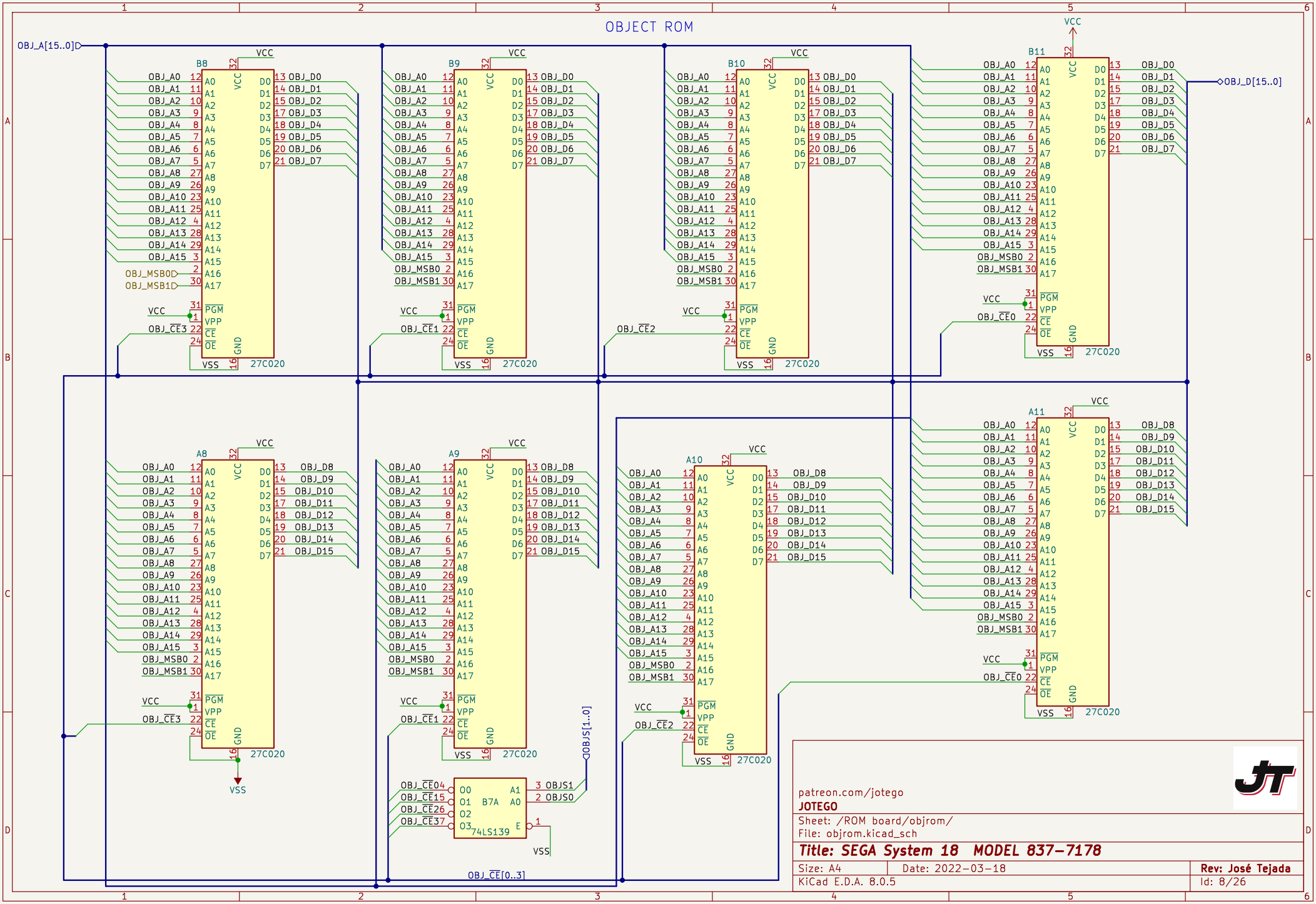
Sheet: /ROM board/scrrom/

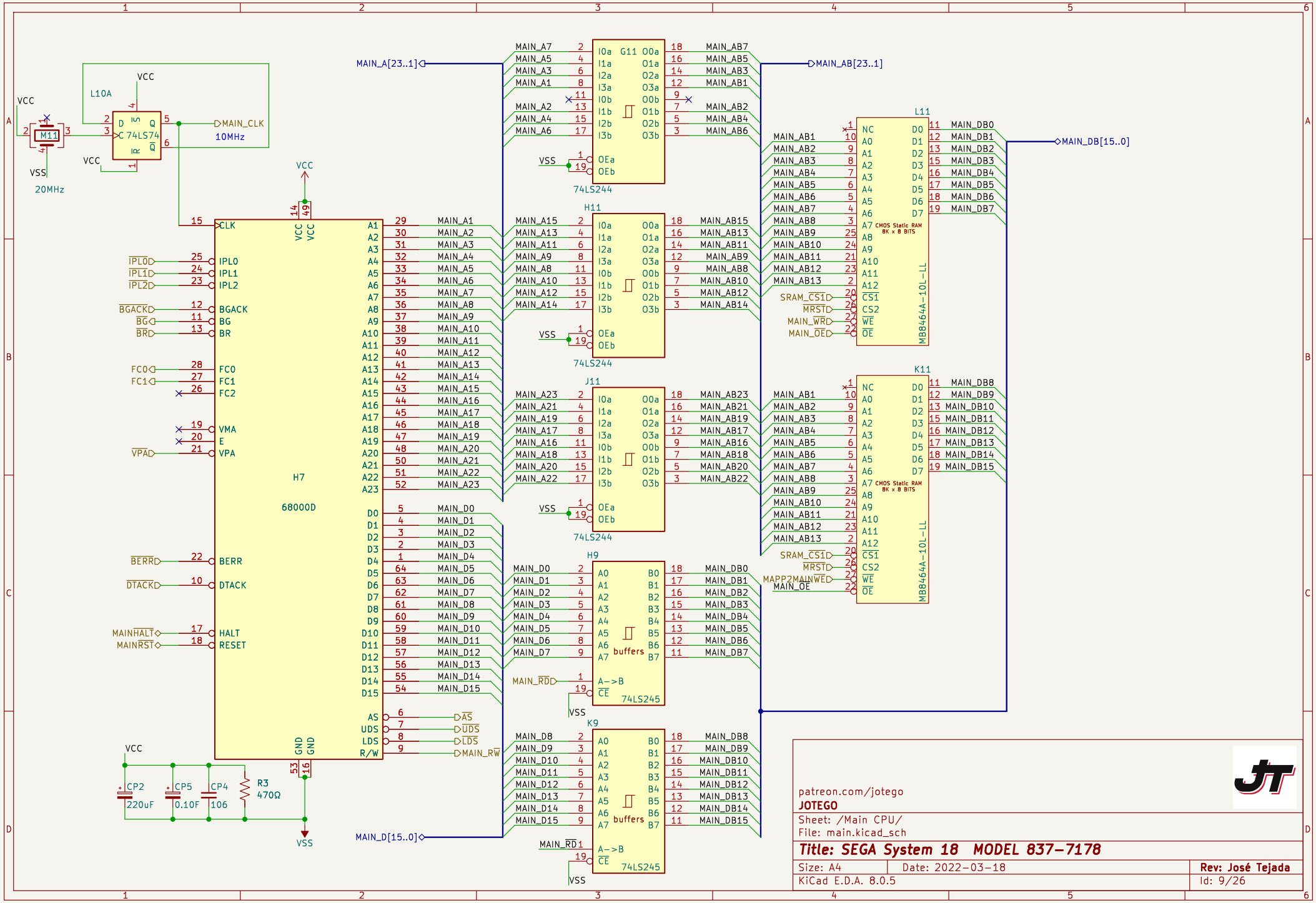
File: scrrom.kicad_sch

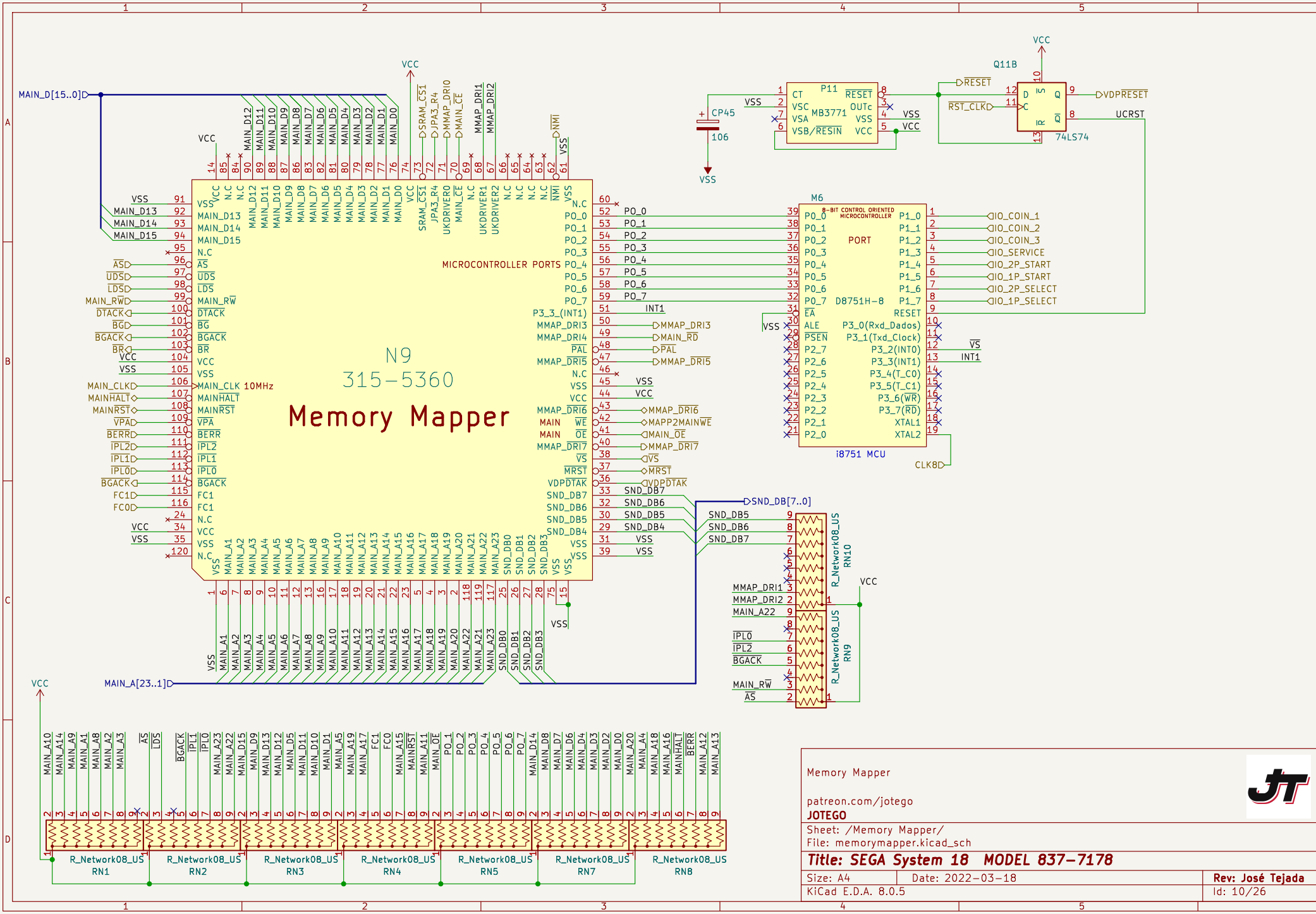
Title: SEGA System 18 MODEL 837-7178

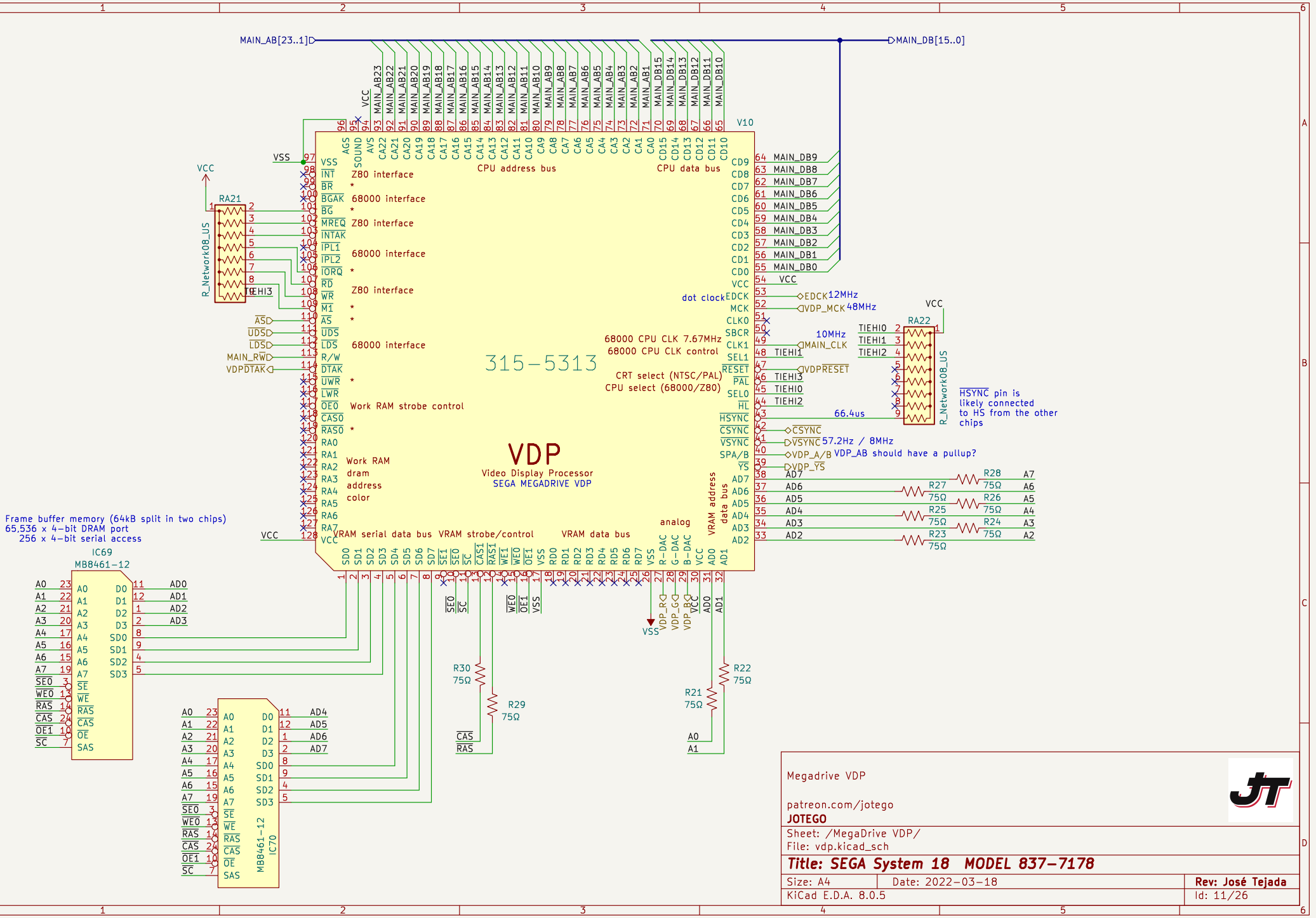
Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 6/26

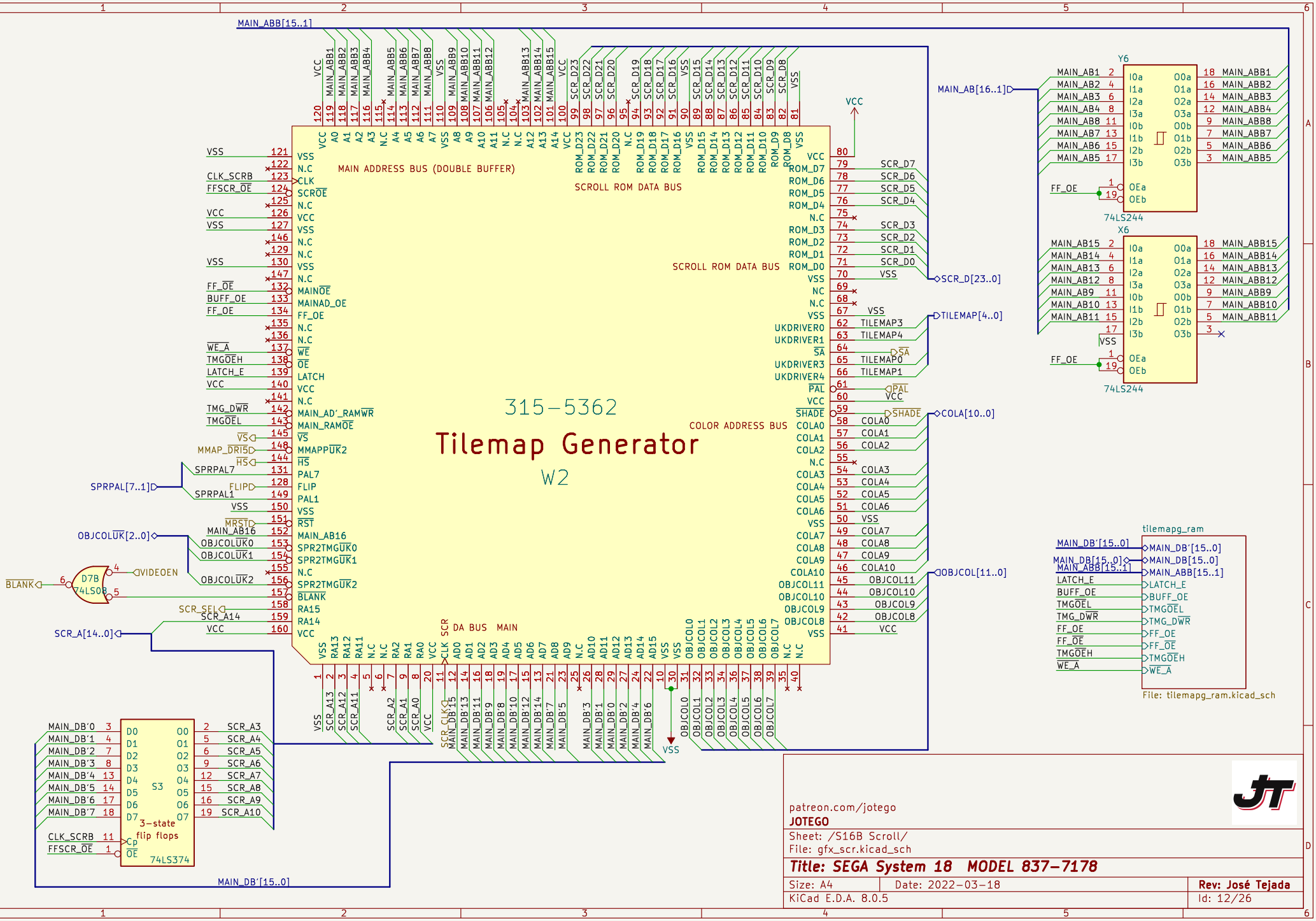

SND_AB[15..0]
A4
B4
B5
B6
27C010
27C020
OTP EPROM 256 K x 8 Bit
SND_DB[7..0]
PCM_SNDROM
PCM_CE0
PCM_CE1
PCM_CE2
MAIN_AB[18..1]
A5
A6
LSB <--> MSB
could be swapped!
MAIN_DB[15..0]
MAIN_CE
MAIN_OE
patreon.com/jotego
JOTEGO
Sheet: /ROM board/cpurom/
File: cpurom.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 7/26

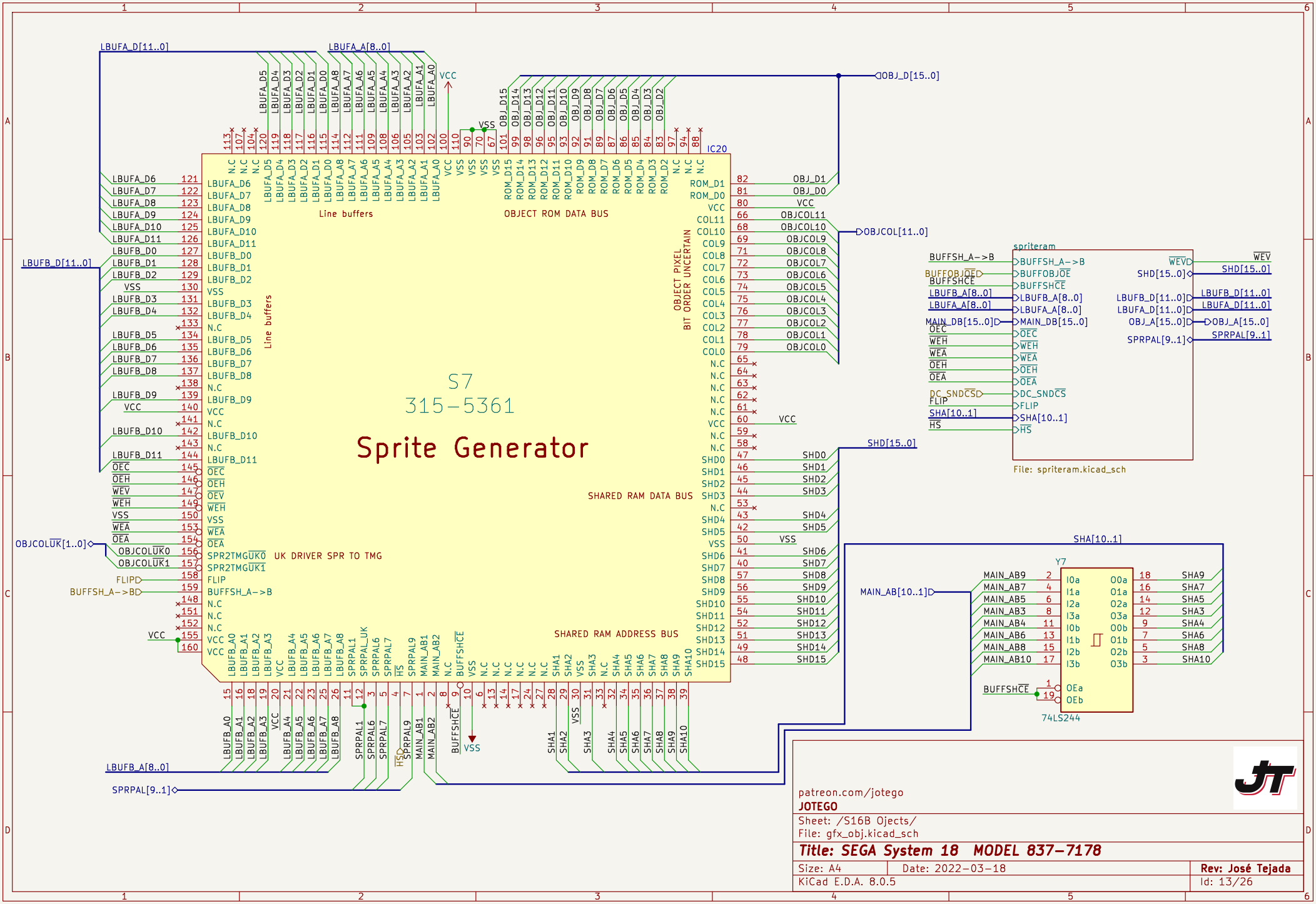

OBJECT ROM
OBJ_A[15..0]
OBJ_D[15..0]
B8
B9
B10
B11
A8
A9
A10
A11
27C020
VCC
VSS
OBJ_MSB0
OBJ_MSB1
OBJ_CE3
OBJ_CE1
OBJ_CE2
OBJ_CE0
OBJS[1..0]
OBJ_CE[0..3]
74LS139
B7A
OBJS1
OBJS0
patreon.com/jotego
JOTEGO
Sheet: /ROM board/objrom/
File: objrom.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 8.0.5
Rev: José Tejada
Id: 8/26

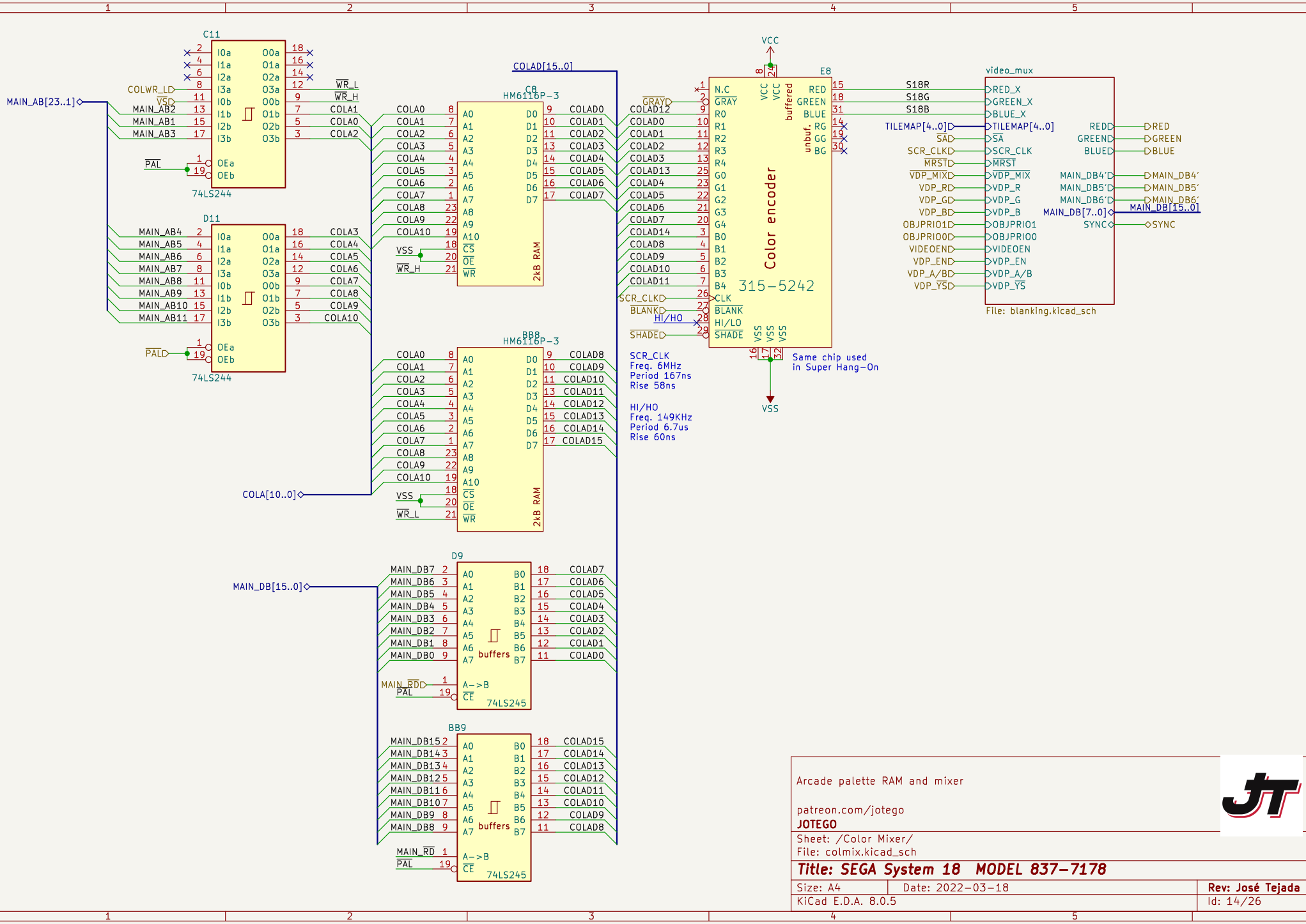

VCC
L10A
74LS74
M11
20MHz
MAIN_CLK
10MHz
VSS
H7
68000D
CLK
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
MAINHALT
MAINRST
AS
UDS
LDS
R/W
MAIN_RW
MAIN_A[23..1]
MAIN_D[15..0]
MAIN_AB[23..1]
MAIN_DB[15..0]
74LS244
74LS245
buffers
H11
J11
H9
K9
L11
K11
MB8464A-10L-LL
CMOS Static RAM 8K x 8 BITS
SRAM_CS1
MRST
MAIN_WR
MAIN_OE
MAPP2MAINWE
CP2
220uF
CP5
0.10F
CP4
106
R3
470Ω
patreon.com/jotego
JOTEGO
Sheet: /Main CPU/
File: main.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 8.0.5
Rev: José Tejada
Id: 9/26

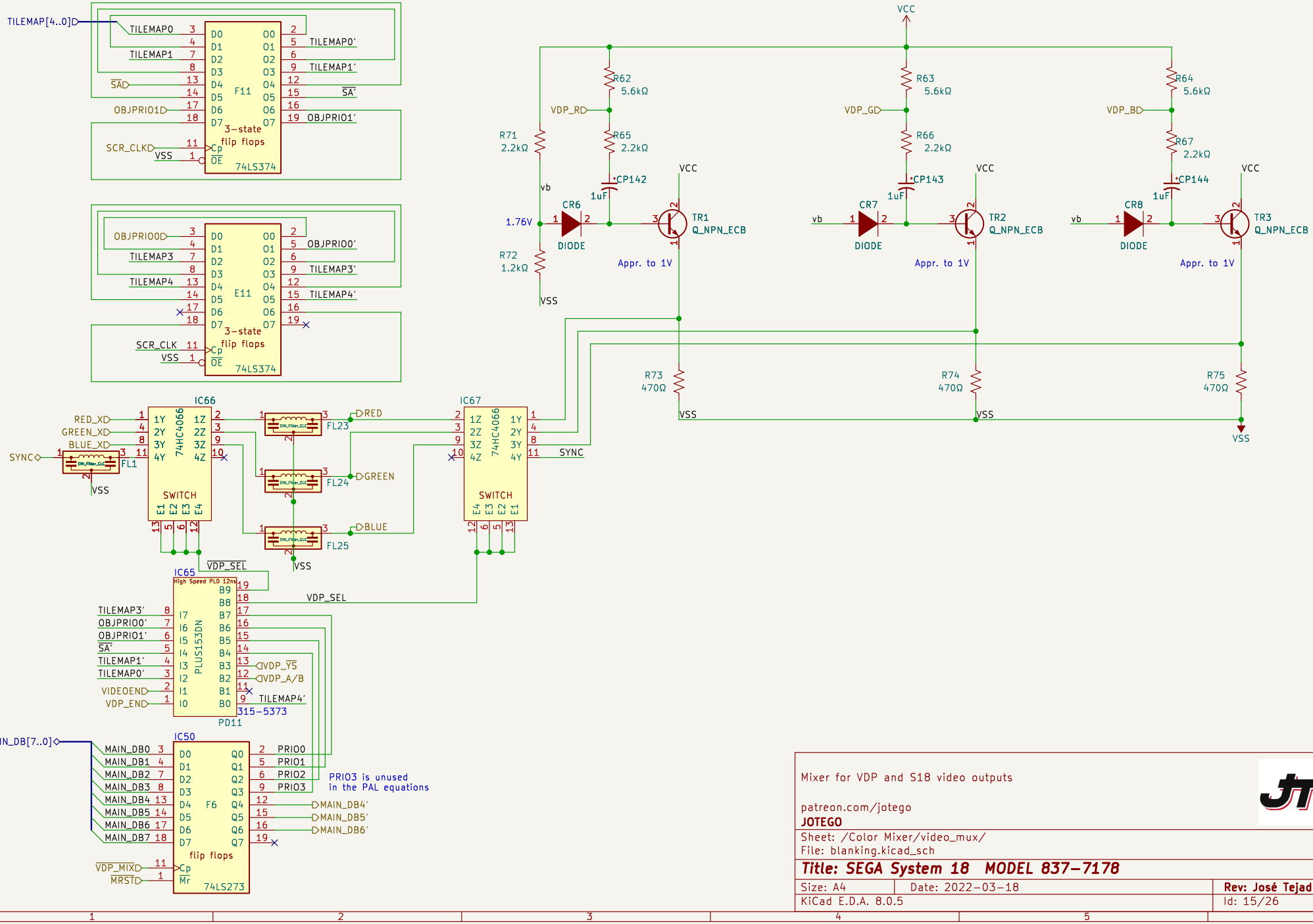

N9
315-5360
Memory Mapper
MICROCONTROLLER PORTS
MAIN_D[15..0]
MAIN_A[23..1]
SND_DB[7..0]
MAIN_CLK 10MHz
M6
8-BIT CONTROL ORIENTED MICROCONTROLLER
PORT
D8751H-8
i8751 MCU
Q11B
74LS74
P11
MB3771
CP45
106
R_Network08_US
RN1
RN2
RN3
RN4
RN5
RN7
RN8
RN9
RN10
Memory Mapper
patreon.com/jotego
JOTEGO
Sheet: /Memory Mapper/
File: memorymapper.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 10/26

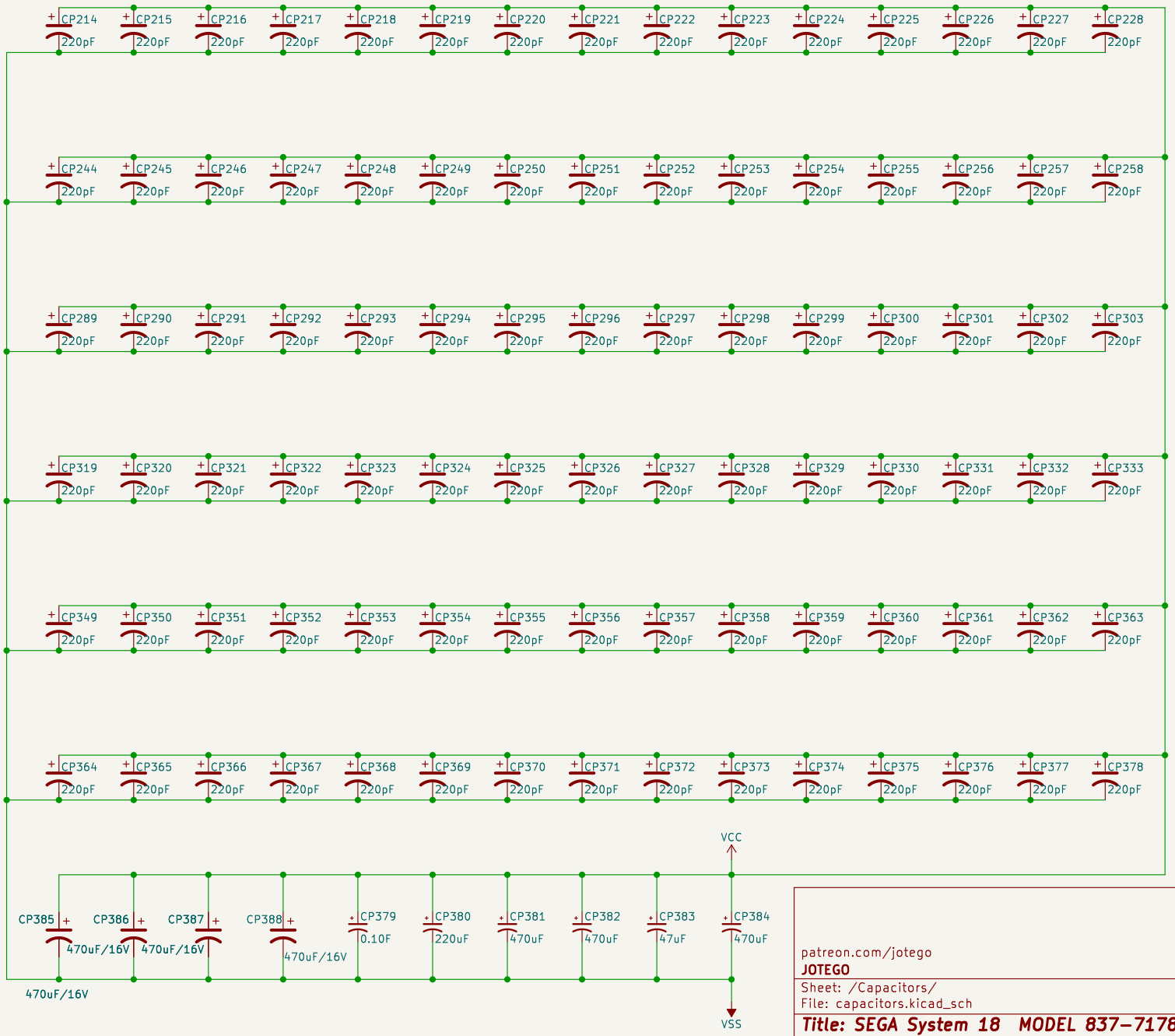

MAIN_AB[23..1]
MAIN_DB[15..0]
VDP
Video Display Processor
SEGA MEGADRIVE VDP
315-5313
CPU address bus
CPU data bus
Z80 interface
68000 interface
Work RAM strobe control
Work RAM dram address color
VRAM serial data bus
VRAM strobe/control
VRAM data bus
VRAM address data bus
analog
68000 CPU CLK 7.67MHz
68000 CPU CLK control
CRT select (NTSC/PAL)
CPU select (68000/Z80)
dot clock
12MHz
48MHz
10MHz
66.4us
57.2Hz / 8MHz
VDP_AB should have a pullup?
HSYNC pin is likely connected to HS from the other chips
Frame buffer memory (64kB split in two chips)
65,536 x 4-bit DRAM port
256 x 4-bit serial access
IC69
MB8461-12
IC70
R_Network08_US
RA21
RA22
Megadrive VDP
patreon.com/jotego
JOTEGO
Sheet: /MegaDrive VDP/
File: vdp.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 11/26

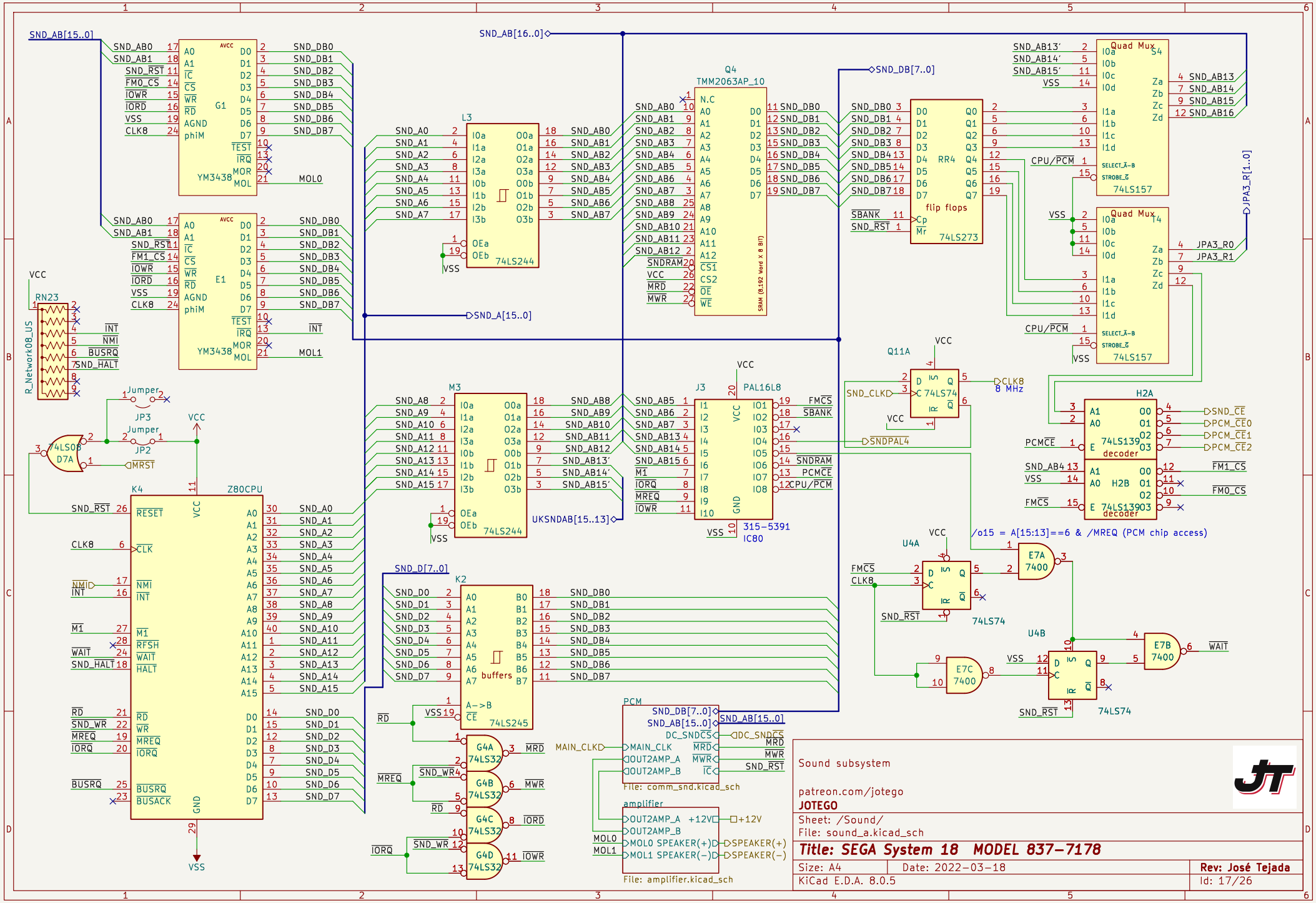

315-5362
Tilemap Generator
W2
MAIN ADDRESS BUS (DOUBLE BUFFER)
SCROLL ROM DATA BUS
COLOR ADDRESS BUS
MAIN_ABB[15..1]
MAIN_AB[16..1]
SCR_D[23..0]
TILEMAP[4..0]
COLA[10..0]
OBJCOL[11..0]
SPRPAL[7..1]
OBJCOLUK[2..0]
SCR_A[14..0]
MAIN_DB'[15..0]
Y6
X6
74LS244
tilemapg_ram
File: tilemapg_ram.kicad_sch
S3
3-state
flip flops
74LS374
patreon.com/jotego
JOTEGO
Sheet: /S16B Scroll/
File: gfx_scr.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 12/26

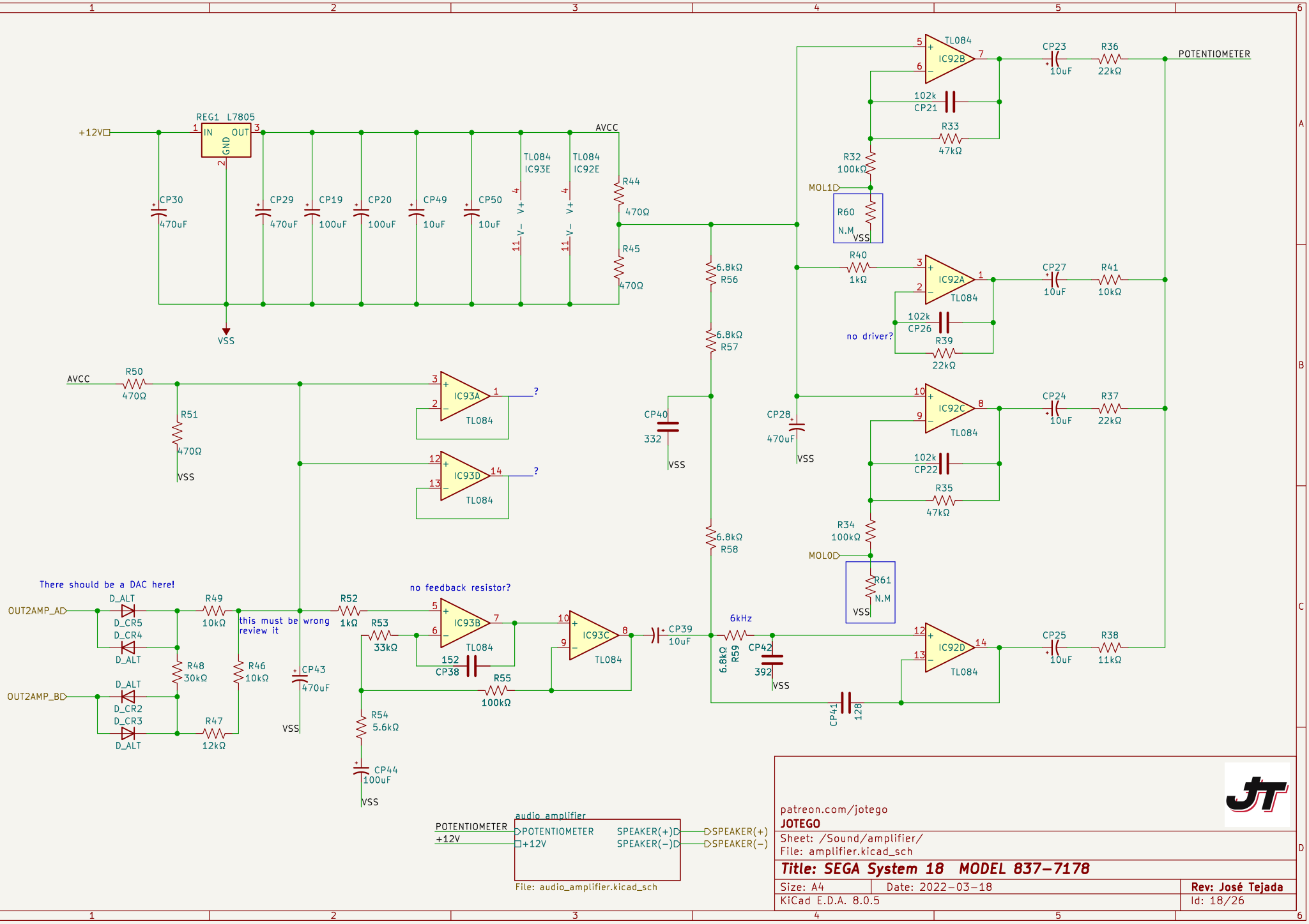

S7
315-5361
Sprite Generator
IC20
Line buffers
OBJECT ROM DATA BUS
OBJECT PIXEL
BIT ORDER UNCERTAIN
SHARED RAM DATA BUS
SHARED RAM ADDRESS BUS
UK DRIVER SPR TO TMG
LBUFA_D[11..0]
LBUFA_A[8..0]
LBUFB_D[11..0]
LBUFB_A[8..0]
OBJ_D[15..0]
OBJCOL[11..0]
SHD[15..0]
OBJCOLUK[1..0]
SHA[10..1]
SPRPAL[9..1]
VCC
VSS
FLIP
BUFFSH_A->B
MAIN_AB[10..1]
spriteram
File: spriteram.kicad_sch
Y7
74LS244
patreon.com/jotego
JOTEGO
Sheet: /S16B Ojects/
File: gfx_obj.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 13/26


C11
74LS244
D11
74LS244
C8
HM6116P-3
2kB RAM
BB8
HM6116P-3
2kB RAM
D9
74LS245
BB9
74LS245
E8
Color encoder
315-5242
video_mux
File: blanking.kicad_sch
COLAD[15..0]
COLA[10..0]
MAIN_AB[23..1]
MAIN_DB[15..0]
SCR_CLK
Freq. 6MHz
Period 167ns
Rise 58ns
HI/HO
Freq. 149KHz
Period 6.7us
Rise 60ns
Same chip used
in Super Hang-On
Arcade palette RAM and mixer
patreon.com/jotego
JOTEGO
Sheet: /Color Mixer/
File: colmix.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 8.0.5
Rev: José Tejada
Id: 14/26

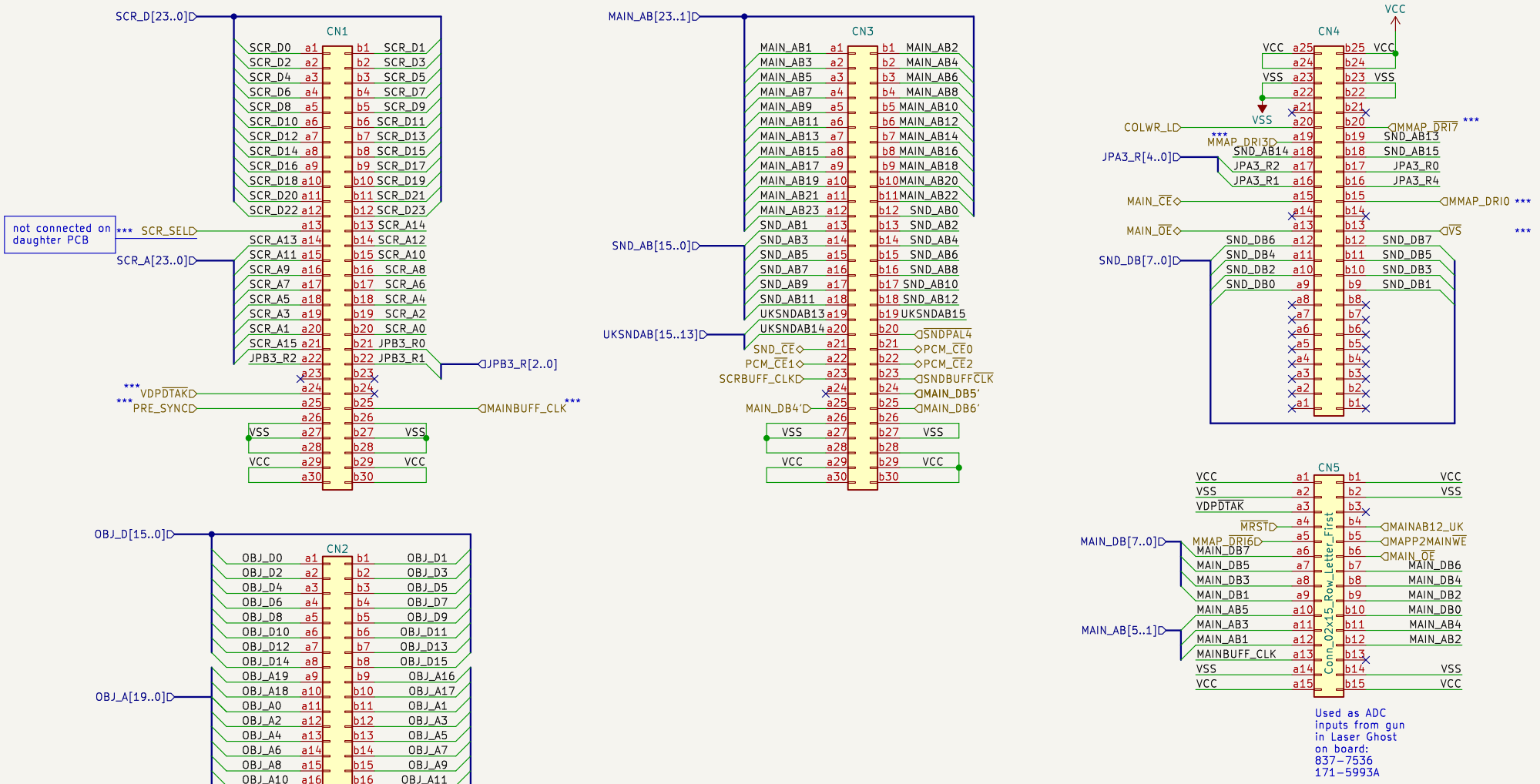

TILEMAP[4..0]
F11
3-state flip flops
74LS374
E11
OBJPRIO0
OBJPRIO1
TILEMAP0
TILEMAP1
TILEMAP3
TILEMAP4
SA
SCR_CLK
VSS
VCC
R62 5.6kΩ
R63 5.6kΩ
R64 5.6kΩ
R65 2.2kΩ
R66 2.2kΩ
R67 2.2kΩ
R71 2.2kΩ
R72 1.2kΩ
VDP_RD
VDP_GD
VDP_BD
CP142 1uF
CP143 1uF
CP144 1uF
CR6
CR7
CR8
DIODE
1.76V
vb
TR1 Q_NPN_ECB
TR2 Q_NPN_ECB
TR3 Q_NPN_ECB
Appr. to 1V
R73 470Ω
R74 470Ω
R75 470Ω
IC66
IC67
74HC4066
SWITCH
RED_X
GREEN_X
BLUE_X
SYNC
FL1
FL23
FL24
FL25
RED
GREEN
BLUE
IC65
PLUS153DN
High Speed PLD 12ns
315-5373
PD11
VDP_SEL
VDP_YS
VDP_A/B
TILEMAP4'
VIDEOEN
VDP_EN
IC50
MAIN_DB[7..0]
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
PRIO0
PRIO1
PRIO2
PRIO3
PRIO3 is unused in the PAL equations
MAIN_DB4'
MAIN_DB5'
MAIN_DB6'
VDP_MIX
MRST
74LS273
Mixer for VDP and S18 video outputs
patreon.com/jotego
JOTEGO
Sheet: /Color Mixer/video_mux/
File: blanking.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 15/26

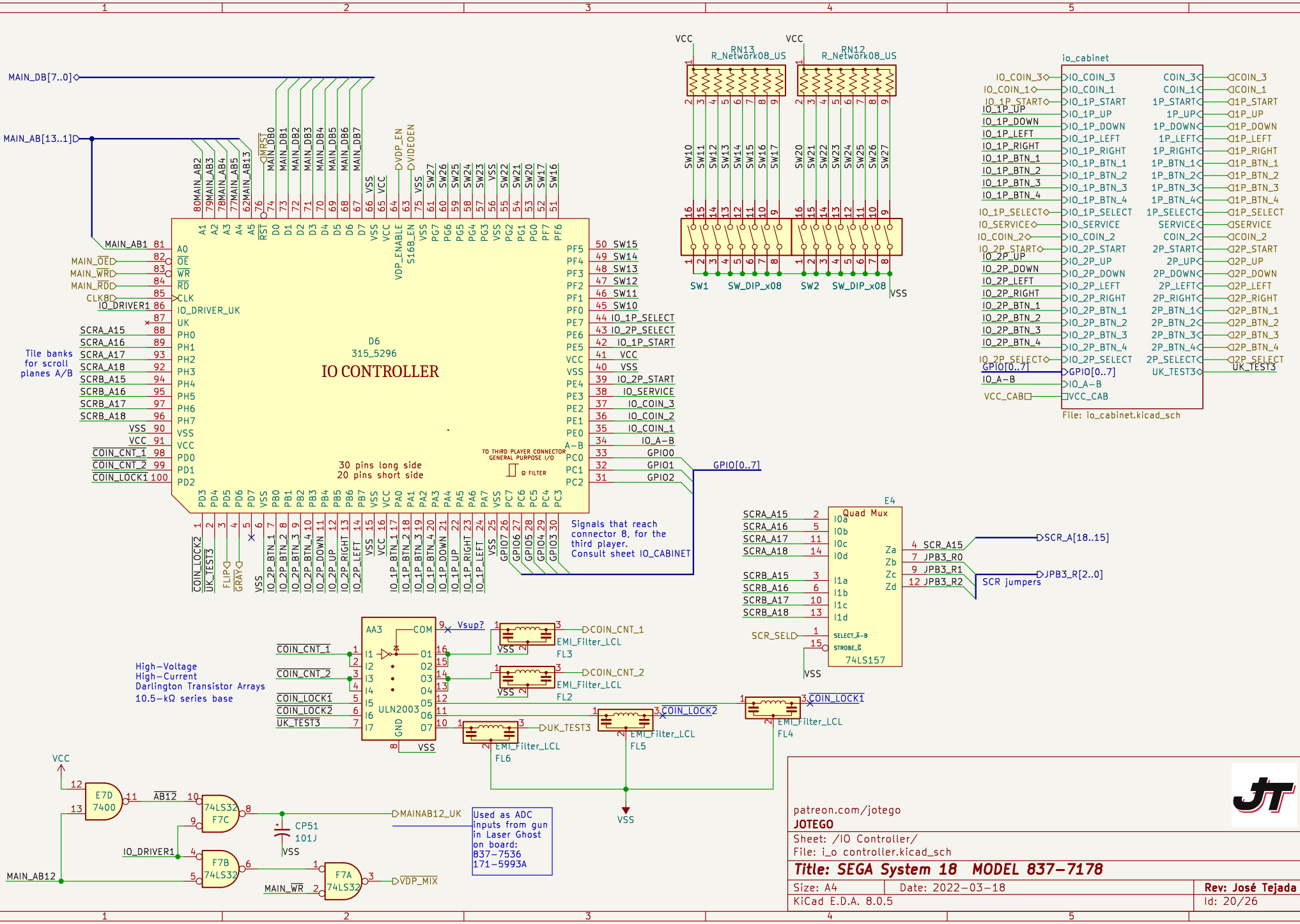

CP214 220pF
CP215 220pF
CP216 220pF
CP217 220pF
CP218 220pF
CP219 220pF
CP220 220pF
CP221 220pF
CP222 220pF
CP223 220pF
CP224 220pF
CP225 220pF
CP226 220pF
CP227 220pF
CP228 220pF
CP244 220pF
CP245 220pF
CP246 220pF
CP247 220pF
CP248 220pF
CP249 220pF
CP250 220pF
CP251 220pF
CP252 220pF
CP253 220pF
CP254 220pF
CP255 220pF
CP256 220pF
CP257 220pF
CP258 220pF
CP289 220pF
CP290 220pF
CP291 220pF
CP292 220pF
CP293 220pF
CP294 220pF
CP295 220pF
CP296 220pF
CP297 220pF
CP298 220pF
CP299 220pF
CP300 220pF
CP301 220pF
CP302 220pF
CP303 220pF
CP319 220pF
CP320 220pF
CP321 220pF
CP322 220pF
CP323 220pF
CP324 220pF
CP325 220pF
CP326 220pF
CP327 220pF
CP328 220pF
CP329 220pF
CP330 220pF
CP331 220pF
CP332 220pF
CP333 220pF
CP349 220pF
CP350 220pF
CP351 220pF
CP352 220pF
CP353 220pF
CP354 220pF
CP355 220pF
CP356 220pF
CP357 220pF
CP358 220pF
CP359 220pF
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CP366 220pF
CP367 220pF
CP368 220pF
CP369 220pF
CP370 220pF
CP371 220pF
CP372 220pF
CP373 220pF
CP374 220pF
CP375 220pF
CP376 220pF
CP377 220pF
CP378 220pF
VCC
CP385 470uF/16V
CP386 470uF/16V
CP387
CP388 470uF/16V
CP379 0.10F
CP380 220uF
CP381 470uF
CP382 470uF
CP383 47uF
CP384 470uF
470uF/16V
VSS
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JOTEGO
Sheet: /Capacitors/
File: capacitors.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 16/26

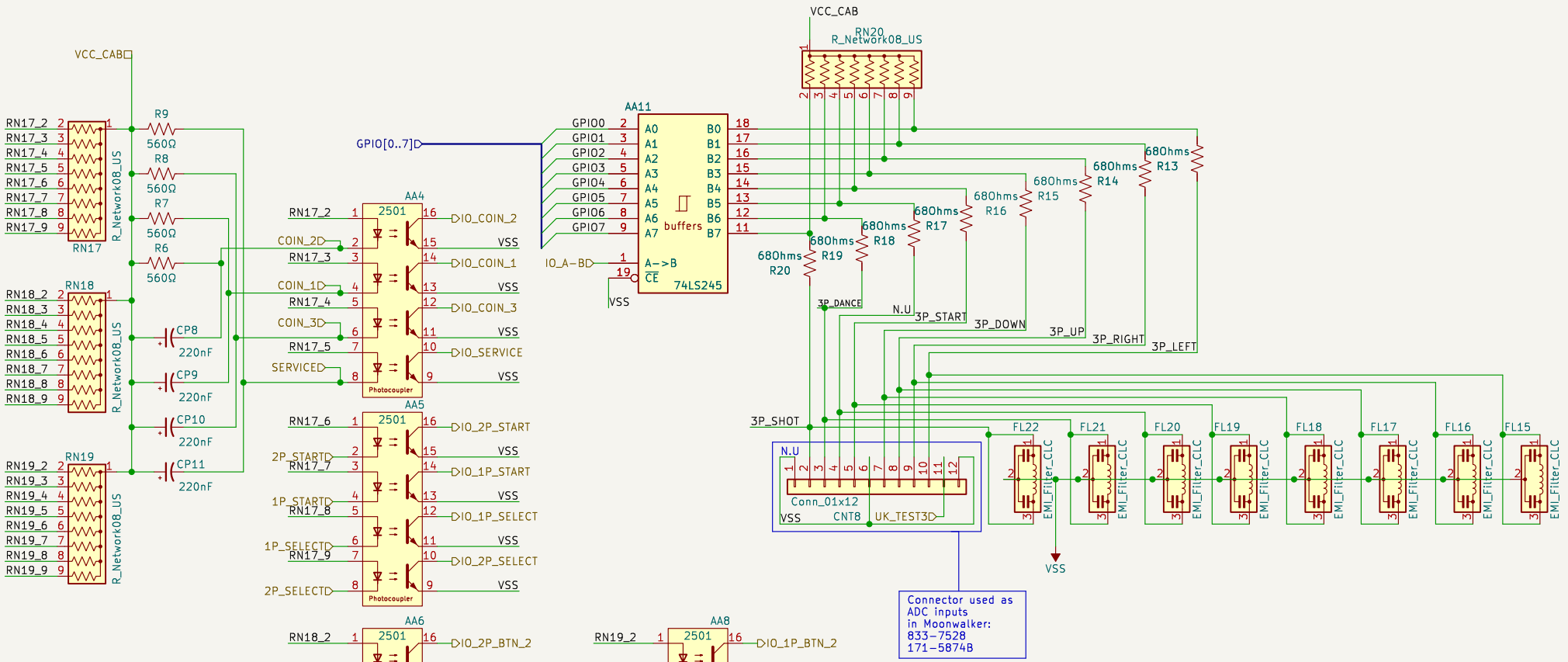

SND_AB[15..0]
SND_AB[16..0]
SND_A[15..0]
SND_D[7..0]
SND_DB[7..0]
UKSNDAB[15..13]
JPA3_R[1..0]
G1
E1
YM3438
L3
74LS244
Q4
TMM2063AP_10
SRAM (8:192 Word X 8 Bit)
flip flops
74LS273
Quad Mux
74LS157
RR4
R_Network08_US
RN23
JP3
JP2
Jumper
74LS08
K4
Z80CPU
M3
J3
PAL16L8
315-5391
IC80
K2
buffers
74LS245
Q11A
74LS74
H2A
74LS139
decoder
U4A
U4B
E7A
E7B
E7C
7400
/o15 = A[15:13]==6 & /MREQ (PCM chip access)
PCM
File: comm_snd.kicad_sch
amplifier
File: amplifier.kicad_sch
Sound subsystem
patreon.com/jotego
JOTEGO
Sheet: /Sound/
File: sound_a.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
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REG1 L7805
+12V
IN
OUT
GND
VSS
AVCC
CP30 470uF
CP29 470uF
CP19 100uF
CP20 100uF
CP49 10uF
CP50 10uF
TL084 IC93E
TL084 IC92E
R44 470Ω
R45 470Ω
R50 470Ω
R51 470Ω
IC93A
IC93D
TL084
?
There should be a DAC here!
OUT2AMP_AD
OUT2AMP_BD
D_ALT
D_CR5
D_CR4
D_CR2
D_CR3
R49 10kΩ
R48 30kΩ
R46 10kΩ
R47 12kΩ
this must be review it
wrong
R52 1kΩ
CP43 470uF
R53 33kΩ
no feedback resistor?
IC93B
IC93C
CP38 152
R55 100kΩ
R54 5.6kΩ
CP44 100uF
CP39 10uF
6kHz
CP42 392
CP41 128
R56 6.8kΩ
R57 6.8kΩ
R58 6.8kΩ
R59 6.8kΩ
CP40 332
CP28 470uF
TL084 IC92B
CP23 10uF
R36 22kΩ
POTENTIOMETER
102k CP21
R33 47kΩ
R32 100kΩ
MOL1
R60 N.M
R40 1kΩ
IC92A
CP27 10uF
R41 10kΩ
102k CP26
no driver?
R39 22kΩ
IC92C
CP24 10uF
R37 22kΩ
102k CP22
R35 47kΩ
R34 100kΩ
MOL0
R61 N.M
IC92D
CP25 10uF
R38 11kΩ
audio_amplifier
POTENTIOMETER
+12V
SPEAKER(+)
SPEAKER(−)
File: audio_amplifier.kicad_sch
patreon.com/jotego
JOTEGO
Sheet: /Sound/amplifier/
File: amplifier.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 8.0.5
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MOTHER AND DAUGHTER PCB CONNECTORS




MAIN_DB[7..0]
MAIN_AB[13..1]
D6
315_5296
IO CONTROLLER
30 pins long side
20 pins short side
Tile banks for scroll planes A/B
TO THIRD PLAYER CONNECTOR GENERAL PURPOSE I/O
Q FILTER
GPIO[0..7]
Signals that reach connector 8, for the third player. Consult sheet IO_CABINET
RN13 R_Network08_US
RN12 R_Network08_US
SW1 SW_DIP_x08
SW2 SW_DIP_x08
io_cabinet
File: io_cabinet.kicad_sch
E4 Quad Mux 74LS157
SCR_A[18..15]
JPB3_R[2..0]
SCR jumpers
AA3 ULN2003
High-Voltage High-Current Darlington Transistor Arrays 10.5-kΩ series base
EMI_Filter_LCL FL3
EMI_Filter_LCL FL2
EMI_Filter_LCL FL4
EMI_Filter_LCL FL5
EMI_Filter_LCL FL6
E7D 7400
74LS32 F7C
F7B 74LS32
F7A 74LS32
CP51 101J
MAINAB12_UK
Used as ADC inputs from gun in Laser Ghost on board: 837-7536 171-5993A
VDP_MIX
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JOTEGO
Sheet: /IO Controller/
File: i_o controller.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 20/26

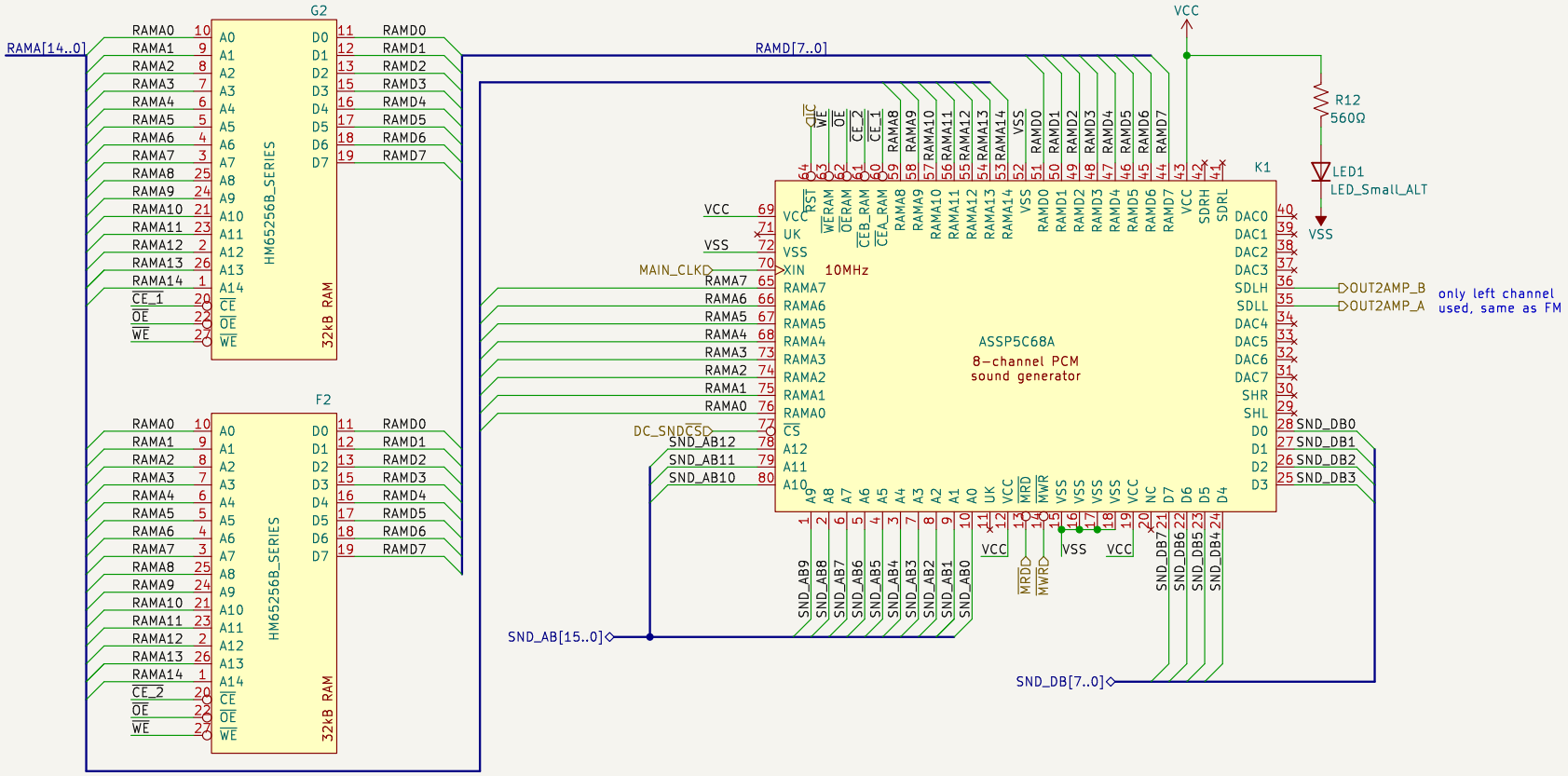


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Sheet: /IO Controller/io_cabinet/
File: io_cabinet.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 21/26



PCM sound generator

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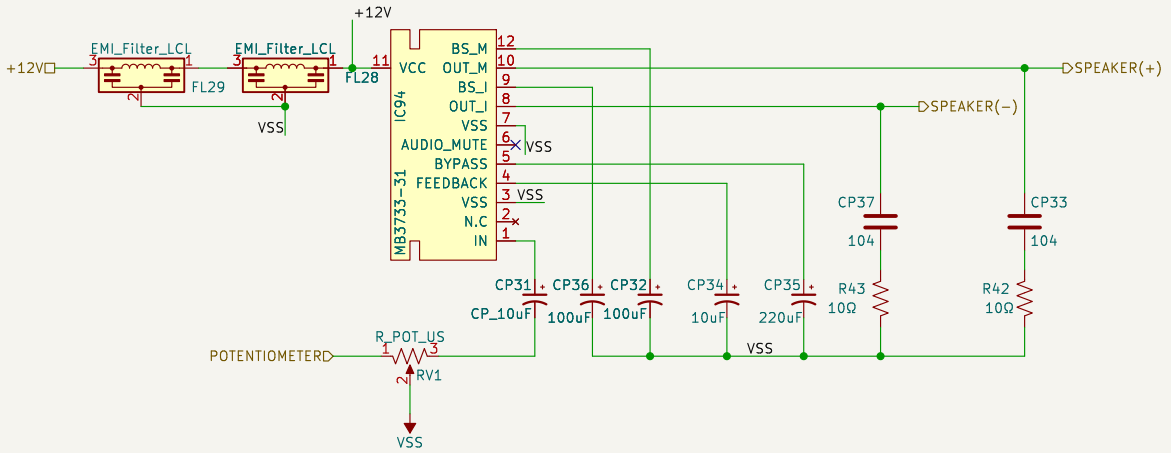
JOTEGO

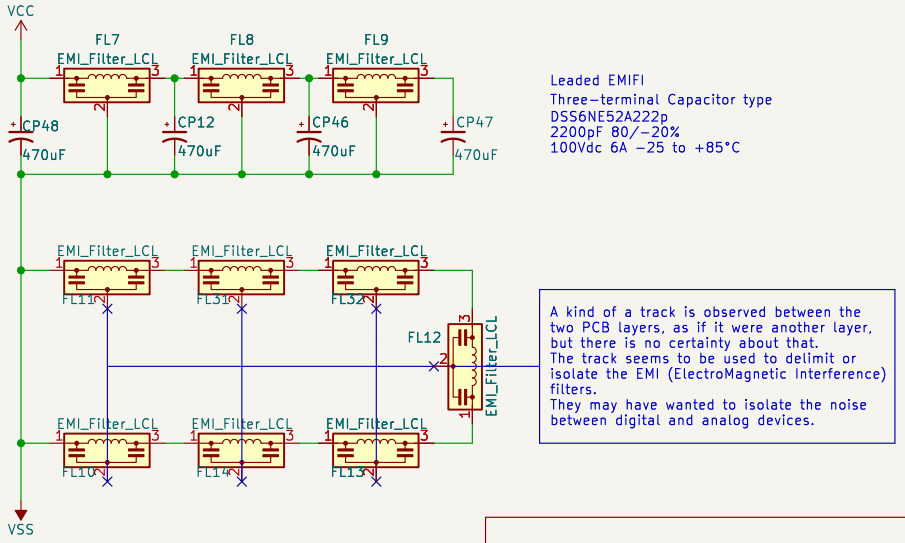
Sheet: /Sound/PCM/

File: comm_snd.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 22/26





Leaded EMIFI
Three-terminal Capacitor type
DSS6NE52A222p
2200pF 80/-20%
100Vdc 6A -25 to +85°C

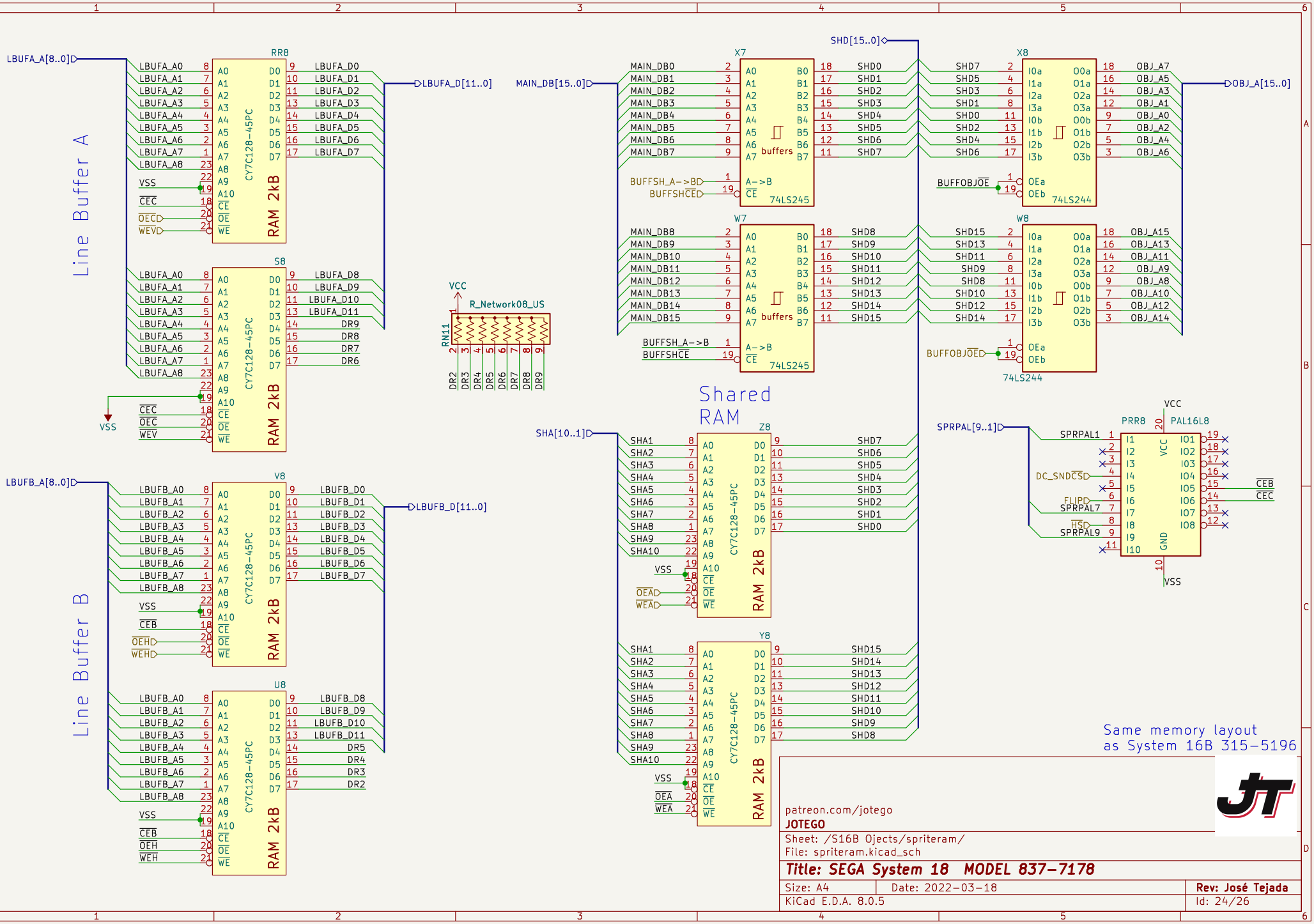
A kind of a track is observed between the two PCB layers, as if it were another layer, but there is no certainty about that.
The track seems to be used to delimit or isolate the EMI (ElectroMagnetic Interference) filters.
They may have wanted to isolate the noise between digital and analog devices.

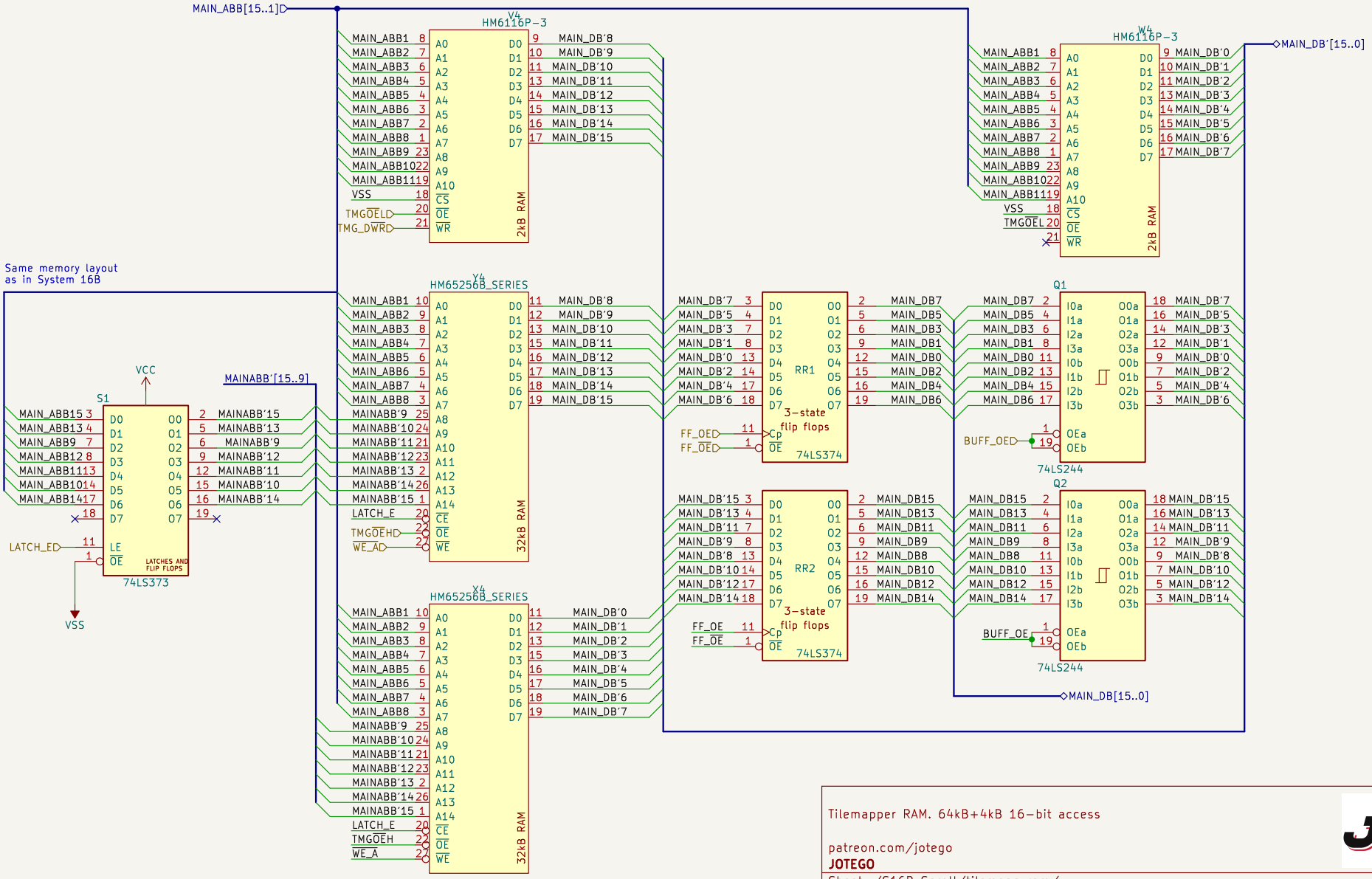
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JOTEGO

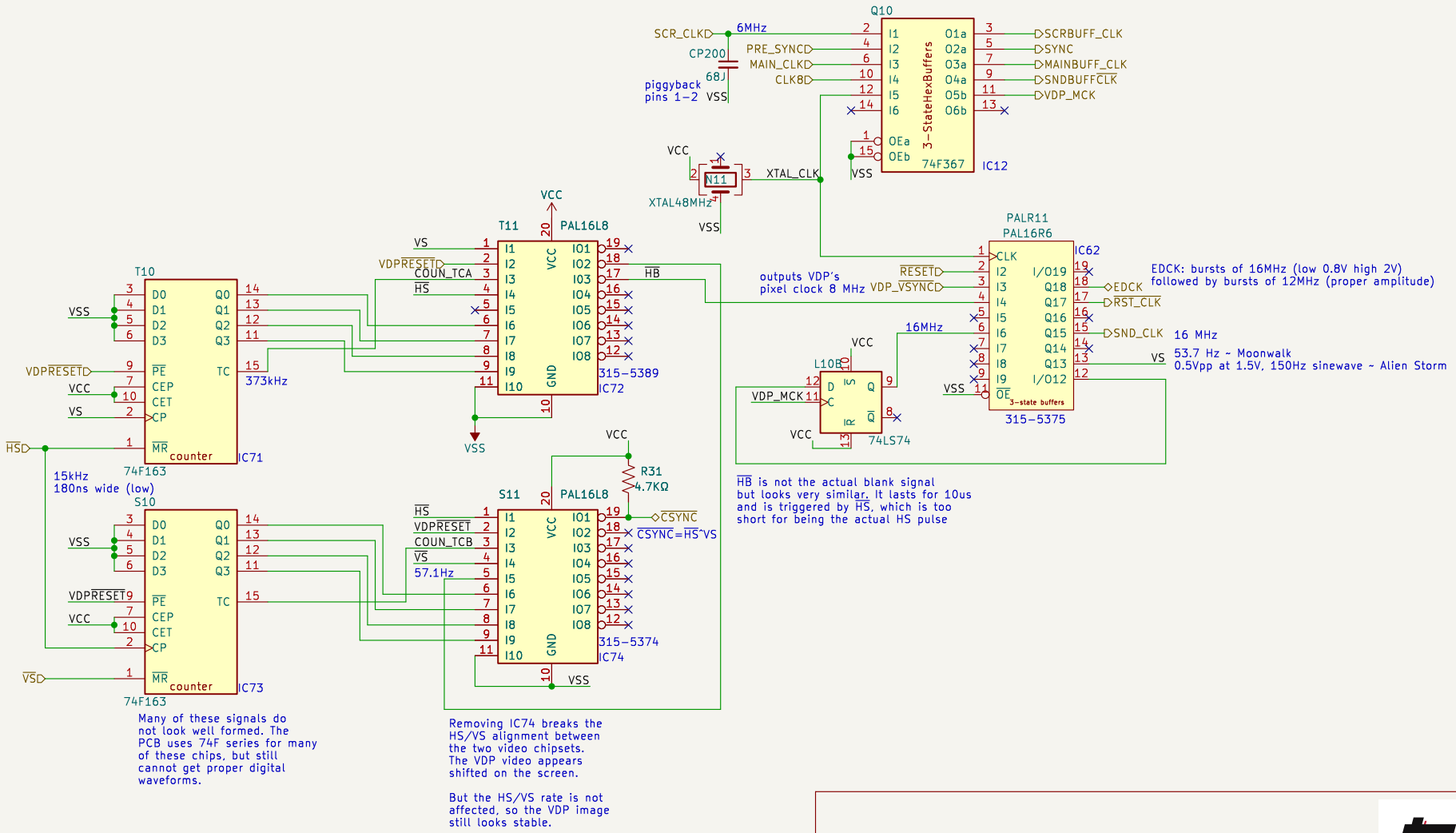
Sheet: /Sound/amplifier/audio_amplifier/
File: audio_amplifier.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 23/26


Line Buffer A
Line Buffer B
Shared RAM
RR8
S8
V8
U8
X7
W7
X8
W8
Z8
Y8
RN11
PRR8
CY7C128-45PC
RAM 2kB
R_Network08_US
74LS245
74LS244
buffers
PAL16L8
LBUFA_A[8..0]
LBUFA_D[11..0]
LBUFB_A[8..0]
LBUFB_D[11..0]
MAIN_DB[15..0]
SHD[15..0]
SHA[10..1]
OBJ_A[15..0]
SPRPAL[9..1]
BUFFSH_A->B
BUFFSHCE
BUFFOBJOE
DC_SNDCS
FLIP
HS
Same memory layout
as System 16B 315-5196
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JOTEGO
Sheet: /S16B Ojects/spriteram/
File: spriteram.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 24/26


MAIN_ABB[15..1]
V4
HM6116P-3
MAIN_ABB1 8
MAIN_ABB2 7
MAIN_ABB3 6
MAIN_ABB4 5
MAIN_ABB5 4
MAIN_ABB6 3
MAIN_ABB7 2
MAIN_ABB8 1
MAIN_ABB9 23
MAIN_ABB10 22
MAIN_ABB11 19
VSS 18
TMGOELD 20
TMG_DWRD 21
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10
CS OE WR
D0 D1 D2 D3 D4 D5 D6 D7
2kB RAM
MAIN_DB'8 9
MAIN_DB'9 10
MAIN_DB'10 11
MAIN_DB'11 13
MAIN_DB'12 14
MAIN_DB'13 15
MAIN_DB'14 16
MAIN_DB'15 17
W4
HM6116P-3
MAIN_ABB1 8
MAIN_ABB2 7
MAIN_ABB3 6
MAIN_ABB4 5
MAIN_ABB5 4
MAIN_ABB6 3
MAIN_ABB7 2
MAIN_ABB8 1
MAIN_ABB9 23
MAIN_ABB10 22
MAIN_ABB11 19
VSS 18
TMGOEL 20
21
2kB RAM
MAIN_DB'0 9
MAIN_DB'1 10
MAIN_DB'2 11
MAIN_DB'3 13
MAIN_DB'4 14
MAIN_DB'5 15
MAIN_DB'6 16
MAIN_DB'7 17
MAIN_DB'[15..0]
Same memory layout
as in System 16B
VCC
MAINABB'[15..9]
S1
MAIN_ABB15 3
MAIN_ABB13 4
MAIN_ABB9 7
MAIN_ABB12 8
MAIN_ABB11 13
MAIN_ABB10 14
MAIN_ABB14 17
18
LATCH_ED 11
1
LE
OE
LATCHES AND FLIP FLOPS
74LS373
VSS
D0 D1 D2 D3 D4 D5 D6 D7
O0 O1 O2 O3 O4 O5 O6 O7
2 MAINABB'15
5 MAINABB'13
6 MAINABB'9
9 MAINABB'12
12 MAINABB'11
15 MAINABB'10
16 MAINABB'14
19
V4
HM65256B_SERIES
MAIN_ABB1 10
MAIN_ABB2 9
MAIN_ABB3 8
MAIN_ABB4 7
MAIN_ABB5 6
MAIN_ABB6 5
MAIN_ABB7 4
MAIN_ABB8 3
MAINABB'9 25
MAINABB'10 24
MAINABB'11 21
MAINABB'12 23
MAINABB'13 2
MAINABB'14 26
MAINABB'15 1
LATCH_E 20
TMGOEHD 22
WE_AD 27
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14
CE OE WE
D0 D1 D2 D3 D4 D5 D6 D7
32kB RAM
MAIN_DB'8 11
MAIN_DB'9 12
MAIN_DB'10 13
MAIN_DB'11 15
MAIN_DB'12 16
MAIN_DB'13 17
MAIN_DB'14 18
MAIN_DB'15 19
V4
HM65256B_SERIES
MAIN_ABB1 10
MAIN_ABB2 9
MAIN_ABB3 8
MAIN_ABB4 7
MAIN_ABB5 6
MAIN_ABB6 5
MAIN_ABB7 4
MAIN_ABB8 3
MAINABB'9 25
MAINABB'10 24
MAINABB'11 21
MAINABB'12 23
MAINABB'13 2
MAINABB'14 26
MAINABB'15 1
LATCH_E 20
TMGOEH 22
WE_A 27
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14
CE OE WE
D0 D1 D2 D3 D4 D5 D6 D7
32kB RAM
MAIN_DB'0 11
MAIN_DB'1 12
MAIN_DB'2 13
MAIN_DB'3 15
MAIN_DB'4 16
MAIN_DB'5 17
MAIN_DB'6 18
MAIN_DB'7 19
MAIN_DB'7 3
MAIN_DB'5 4
MAIN_DB'3 7
MAIN_DB'1 8
MAIN_DB'0 13
MAIN_DB'2 14
MAIN_DB'4 17
MAIN_DB'6 18
RR1
D0 D1 D2 D3 D4 D5 D6 D7
O0 O1 O2 O3 O4 O5 O6 O7
3-state flip flops
FF_OED 11
FF_OED 1
Cp
OE
74LS374
2 MAIN_DB7
5 MAIN_DB5
6 MAIN_DB3
9 MAIN_DB1
12 MAIN_DB0
15 MAIN_DB2
16 MAIN_DB4
19 MAIN_DB6
MAIN_DB'15 3
MAIN_DB'13 4
MAIN_DB'11 7
MAIN_DB'9 8
MAIN_DB'8 13
MAIN_DB'10 14
MAIN_DB'12 17
MAIN_DB'14 18
RR2
D0 D1 D2 D3 D4 D5 D6 D7
O0 O1 O2 O3 O4 O5 O6 O7
3-state flip flops
FF_OE 11
FF_OE 1
Cp
OE
74LS374
2 MAIN_DB15
5 MAIN_DB13
6 MAIN_DB11
9 MAIN_DB9
12 MAIN_DB8
15 MAIN_DB10
16 MAIN_DB12
19 MAIN_DB14
Q1
MAIN_DB7 2
MAIN_DB5 4
MAIN_DB3 6
MAIN_DB1 8
MAIN_DB0 11
MAIN_DB2 13
MAIN_DB4 15
MAIN_DB6 17
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
18 MAIN_DB'7
16 MAIN_DB'5
14 MAIN_DB'3
12 MAIN_DB'1
9 MAIN_DB'0
7 MAIN_DB'2
5 MAIN_DB'4
3 MAIN_DB'6
BUFF_OED 1
19
OEa
OEb
74LS244
Q2
MAIN_DB15 2
MAIN_DB13 4
MAIN_DB11 6
MAIN_DB9 8
MAIN_DB8 11
MAIN_DB10 13
MAIN_DB12 15
MAIN_DB14 17
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
18 MAIN_DB'15
16 MAIN_DB'13
14 MAIN_DB'11
12 MAIN_DB'9
9 MAIN_DB'8
7 MAIN_DB'10
5 MAIN_DB'12
3 MAIN_DB'14
BUFF_OE 1
19
OEa
OEb
74LS244
MAIN_DB[15..0]
Tilemapper RAM. 64kB+4kB 16-bit access
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JOTEGO
Sheet: /S16B Scroll/tilemapg_ram/
File: tilemapg_ram.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
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Id: 25/26



VDP pixel clock and csync generation

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Sheet: /Video Sync/

File: buffers.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
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